

A Twist on DLA: Decoupling Transport from Aggregation

Candidate Number: 1075289

Hilary Term 2026



*A dissertation submitted for the degree of
Mathematics Part B*

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Chapter 1

Introduction

1.1 Overview

Diffusion Limited Aggregation (DLA) is a stochastic model for irreversible colloidal aggregation. In this model a seed particle is placed at the origin, and then subsequent particles are released from a boundary far from the cluster one at a time. The particle then diffuses, either as a random walk on a lattice or Brownian motion (the continuous analogue to a random walk). Upon collision, the walker sticks and the next one starts diffusing. The cluster, or aggregation, grows in this manner one particle at a time.

The model was first proposed in 1981 by Witten and Sander to model the colloidal aggregation of metal precipitates [1], but has since been found to have applications in a diverse range of physical systems, including electrodeposition [2], bacteria growth [3] and viscous fingering in Hele-Shaw cells [4] among others.

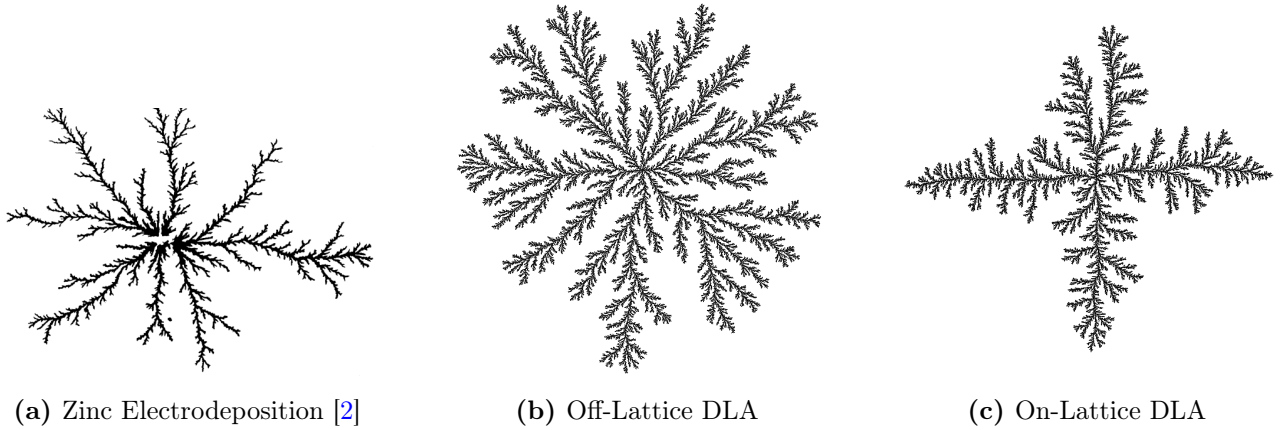


Figure 1.1: Comparison of physical aggregation (a) with the two primary modelling approaches: continuum Off-Lattice DLA (b) and discrete Lattice DLA (c). Both are the results of simulation with 10^7 particles. Note the distinct grid-induced anisotropy (lack of radial symmetry) visible in the Lattice case.

1.2 Link to Potential Theory

While DLA is defined by a stochastic algorithm (random walkers), it can be reframed as a problem within potential theory. A simple symmetric random walk is a process where the probability of a walker being at any given site is the average of the probabilities of being at any of its neighbouring sites. We will show that this results in the steady-state probability density $\phi(u)$ of a diffusing particle at position u satisfying the Laplace equation $\nabla^2\phi = 0$.

In particular, each additional particle that aggregates onto the cluster distorts the cluster's probability field. This compounds over time to cause the field to concentrate at the protruding tips, resulting in the observed complicated structure of DLA clusters [5]. Crucially for this project, this reframing allows us to exploit several potential theoretic results in order to optimise the algorithms we use to generate large clusters.

1.3 Fractal Dimension

DLA is widely recognised as a fractal due to the statistical self-similarity exhibited by its branch-like structure, and is therefore often characterised by its fractal dimension D . This can be thought of as the relationship between the mass of a cluster (number of particles N) and its linear size R , where $N \propto R^D$.

For instance, doubling the size of a line would double its mass, so it has dimension 1, while doubling the size of a solid disc would quadruple its mass, so this has dimension 2. Simulations of Off-Lattice DLA have consistently yielded a fractal dimension $D \approx 1.71$, but this has not been proven [6].

In fact, the only notable mathematical theorem on 2D DLA is due to Kesten, who proved a bound on the growth rate of Lattice DLA in 1987, with a key consequence being that the fractal dimension of a DLA cluster is bounded below by 1.5 [7]. This was generalised by Losev and Smirnov [8] to a wider class of models using a different approach.

1.4 Anisotropy and Thesis Goals

It is likely that the dimension of Lattice DLA is actually lower than Off-Lattice due to the effect of the grid on the cluster shape, a phenomenon known as anisotropy. Grebenkov and Belyaev [9] discuss how anisotropy is likely to reduce the dimension of large enough clusters, but the reasons why, and to what extent this occurs, are still not fully understood.

To answer these questions, it is necessary to compare simulations of both Lattice and Off-Lattice DLA models. Additionally, to understand the effect of the lattice we create a Hybrid algorithm with continuous diffusion but a discrete aggregation method. Other hybrid models of DLA have previously been studied [10], including a similar semi-lattice approach [11], but to our knowledge have not been utilised to decouple the effects of transport from aggregation in Lattice DLA.

The purpose of this dissertation is to compare these three different models of DLA, in order to decouple the diffusion process of DLA from the geometry of the aggregation. By analysing various features of the resulting clusters, we hope to better understand the relationship between the continuous and discrete variations of the probability distribution governing DLA growth.

We begin with a brief outline of the mathematical theory necessary to understand the underlying mechanisms behind DLA in Chapter 2, before explaining how to utilise this theory to create efficient algorithms, including our hybrid approach, in Chapter 3. We briefly outline our simulation results in Chapter 4, then conduct an extended analysis of both the fractal dimension and anisotropy in Chapter 5 and Chapter 6.

Chapter 2

Potential Theory

In this chapter we establish the theoretical framework governing DLA. We assume a foundational understanding of the relevant topics equivalent to all standard Part A courses, as well as the Part B course *Probability, Measure and Martingales* [12]. A brief summary of the prerequisites can be found in Appendix A.

We will establish the link between a simple symmetric random walk (SSRW) (and its continuous analogue, Brownian motion) and the Laplace equation. We will therefore be able to view DLA as a form of moving boundary problem. At each stage, the probability distribution of the first hitting point of the walker on the cluster boundary can be viewed as the solution to a Dirichlet problem.

For large-scale simulations, explicitly solving this Dirichlet problem at each stage is computationally intractable. Therefore, we also present exact solutions to simpler, static boundary problems, which we will exploit in Chapter 3.

2.1 Discrete Potential Theory

We begin with potential theory in a discrete setting, which forms the foundation of our Lattice model. The theoretical framework in this section is largely adapted from Lawler and Limic [5], applied here to the specific case of 2D DLA.

2.1.1 Simple Random Walks and the Generator

Let $(S_n)_{n \geq 0}$ be a stochastic process on $\mathbb{Z}^2$ defined by $S_n = S_0 + \sum_{i=1}^n X_i$, where $X_1, X_2, \dots$ are independent, identically distributed random variables. We are primarily interested in the Simple Symmetric Random Walk (SSRW), where the walker moves to one of its four nearest neighbours with equal probability. We denote this increment distribution as $p(x) = \mathbb{P}(X_i = x)$, where $p(x) = 1/4$ if $|x| = 1$, and 0 otherwise.

The evolution of the expectation of functions under this random walk is governed by a linear operator known as the discrete generator, $\mathcal{L}$. For a function $f : \mathbb{Z}^2 \rightarrow \mathbb{R}$, it is defined probabilistically as the expected change in f during a single step:

$$\mathcal{L}f(x) = \mathbb{E}[f(S_1) - f(S_0) \mid S_0 = x] = \sum_{y \in \mathbb{Z}^2} p(y - x)[f(y) - f(x)].$$

For a SSRW, it is easy to see that this generator is directly proportional to the standard (unnormalised) discrete Laplacian, $\mathcal{L} = \frac{1}{4}\Delta$, where $\Delta f(x) = \sum_{y \sim x} (f(y) - f(x))$, and the notation $y \sim x$ indicates that the sum is taken over all neighbours y of x .

A function f is said to be discrete harmonic on a subset $A \subset \mathbb{Z}^2$ if $\mathcal{L}f(x) = 0$ for all $x \in A$. Equivalently, a harmonic function's value at any site is exactly the average of its nearest neighbours. This specific property guarantees that evaluating a harmonic function along the path of an unbiased random walk yields a martingale.

Proposition 2.1 (Harmonic Functions are Martingales). *Let S_n be a random walk and f be a function harmonic on a set A . Let $\tau_A = \inf\{n \geq 0 : S_n \notin A\}$ be the first exit time from A . Then the stopped process $M_n = f(S_{n \wedge \tau_A})$ is a martingale (with respect to the natural filtration of S_n).*

Proof. See Lawler and Limic [5, Proposition 6.1.1]. □

2.1.2 The Discrete Dirichlet Problem

To model DLA, we must distinguish between the occupied cluster sites, denoted $K \subset \mathbb{Z}^2$, and the available growth sites. We define the boundary ∂K as the set of all empty sites adjacent to the cluster. The active domain for the random walker is therefore the free space excluding this boundary, $\mathbb{Z}^2 \setminus (K \cup \partial K)$.

Consider the probability $u_R(x)$ that a random walker starting at $x \notin K \cup \partial K$ escapes to some large outer bounding box ∂R before hitting the cluster boundary ∂K . Because the process is a SSRW, the probability of escaping from x must be the average of the escape probabilities of its four neighbours. This immediately implies that $u_R(x)$ is a harmonic function, satisfying the discrete Laplace equation $\mathcal{L}u_R(x) = 0$.

Each stage of the DLA process is therefore equivalent to solving the following boundary value problem:

$$\begin{cases} \mathcal{L}u_R(x) = 0 & \text{if } x \notin K \cup \partial K \text{ (Harmonic in free space),} \\ u_R(x) = 0 & \text{if } x \in \partial K \text{ (Hit the cluster boundary),} \\ u_R(x) = 1 & \text{if } x \in \partial R \text{ (Escaped to bounding box).} \end{cases}$$

This is a specific instance of the general discrete Dirichlet problem in 2D.

Theorem 2.2 (Solution to the 2D Discrete Dirichlet Problem). *Let $A \subset \mathbb{Z}^2$ be a subset and let $\tau = \inf\{n \geq 0 : S_n \notin A\}$ be the first exit time. Assume that the walk stops almost surely ($\mathbb{P}_x(\tau < \infty) = 1$ for all $x \in A$) and that the boundary function $g : \partial A \rightarrow \mathbb{R}$ is bounded. Then, the hitting probability function defined by the expectation*

$$f(x) = \mathbb{E}_x[g(S_\tau)] \tag{2.1}$$

is the unique bounded solution to the Dirichlet system $\mathcal{L}f(x) = 0$ for all $x \in A$, subject to $f(x) = g(x)$ for all $x \in \partial A$.

Proof. Existence: If $x \in \partial A$, then $\tau = 0$, so $S_\tau = x$ and $f(x) = \mathbb{E}_x[g(x)] = g(x)$, satisfying the boundary condition. If $x \in A$, we use the Markov property at the first step:

$$f(x) = \mathbb{E}_x[g(S_\tau)] = \sum_{y \sim x} p(y - x) \mathbb{E}_y[g(S_\tau)] = \sum_{y \sim x} p(y - x) f(y).$$

Thus $\mathcal{L}f(x) = 0$.

Uniqueness: Suppose h is another bounded solution to the system. Since h is harmonic, Proposition 2.1 states that the process $M_n = h(S_{n \wedge \tau})$ constitutes a bounded martingale. Given that $\tau < \infty$ almost surely, we can apply the Optional Stopping Theorem (Theorem A.5):

$$h(x) = \mathbb{E}_x[M_0] = \mathbb{E}_x[M_\tau] = \mathbb{E}_x[h(S_\tau)].$$

From the boundary condition, $h(S_\tau) = g(S_\tau)$. Therefore $h(x) = \mathbb{E}_x[g(S_\tau)] = f(x)$, proving uniqueness. $\square$

2.1.3 Recurrence and the Potential Kernel

To define the global aggregation rule for DLA, we must take the limit of the bounding box to infinity ($R \rightarrow \infty$). Because the 2D simple symmetric random walk is recurrent, a walker will return to any finite cluster almost surely. Consequently, as demonstrated by Lawler and Limic [5, Proposition 6.2.2], the probability of escaping to infinity is exactly zero; attempting to solve

the standard Dirichlet problem with 0 on the cluster boundary and 1 at infinity yields only the trivial potential $f \equiv 0$.

To address this, we turn to Green's functions. In the context of differential equations, the Green's function $G(x, y)$ is familiar as the fundamental solution to the Poisson equation $\mathcal{L}G(x, y) = -\delta_y(x)$, representing the potential field at x induced by a point source at y . In the discrete probabilistic setting, this concept translates directly into the expected number of times a random walk starting at x visits the site y :

$$G(x, y) = \sum_{n=0}^{\infty} p_n(y - x).$$

For transient walks ($d \geq 3$), this expected number of visits is finite, and the sum converges. However, the recurrence of 2D walks causes a walker to return to the origin infinitely often, meaning $G(x, y) = \infty$ for all x, y .

We resolve this divergence by regularising the Green's function via potential differences, arriving at the potential kernel.

Definition 2.3 (Potential Kernel). *The potential kernel $a(x)$ for a simple random walk in $\mathbb{Z}^2$ is defined as:*

$$a(x) = \sum_{n=0}^{\infty} [p_n(0) - p_n(x)],$$

where $p_n(x) = \mathbb{P}_0(S_n = x)$.

It is not immediately obvious that this sum is finite. However, Lawler and Limic [5, Section 4.4.1] show that the difference in these transition probabilities can be bounded such that the series converges absolutely. Thus, the potential kernel $a(x)$ is well-defined and finite for all x .

The potential kernel acts as the fundamental solution to the discrete Laplacian. It satisfies the discrete Poisson equation $\mathcal{L}a(x) = \delta_0(x)$ and replaces the 'boundary at infinity' with an asymptotic logarithmic growth condition.

Theorem 2.4 (Asymptotics of the Potential Kernel). *As $|x| \rightarrow \infty$, the potential kernel satisfies:*

$$a(x) = \frac{2}{\pi} \ln |x| + k + O(|x|^{-2}),$$

where k is a constant involving Euler's constant γ .

Proof. See Lawler and Limic [5, Theorem 4.4.4]. □

Remark 2.5. The logarithmic growth may seem arbitrary, but it can be justified with geometric intuition. Consider a circle of radius R centred on the cluster (large enough so that it contains the entire aggregate). For a steady flow of random walkers coming from infinity to reach the cluster, the total flux (rate of flow) across this circle must be independent of R to satisfy the conservation of mass. This total flux is the product of the circumference ($2\pi R$) and the radial gradient of the potential ($\nabla f(R)$). This implies that the gradient must decay as $\nabla f(R) \propto 1/R$. Integrating this gradient yields:

$$f(R) \propto \int \frac{1}{R} dR = \ln R.$$

By transitioning from the trivial probability solution to the regularised potential kernel $a(x)$, we gain a tool that quantifies the relative accessibility of a lattice site from infinity. Because $a(x)$ is finite and strictly well-behaved, it allows us to exactly evaluate the potential field along complex unbounded geometries, such as an infinite line.

2.1.4 Discrete Harmonic Measure

We now possess the tools to formally define the probability distributions governing both the local transport and global aggregation of DLA.

Definition 2.6 (Local Harmonic Measure). *Let $A \subset \mathbb{Z}^2$ be a discrete domain and $x \in A$. The discrete harmonic measure from x , denoted $H_A(x, \cdot)$, is the probability distribution of the first exit point from A . For a boundary point $y \in \partial A$:*

$$H_A(x, y) = \mathbb{P}_x(S_\tau = y),$$

where $\tau = \inf\{n \geq 0 : S_n \notin A\}$ is the first exit time.

This local measure is exactly the hitting probability solved by the discrete Dirichlet problem (Theorem 2.2), and it is the mechanism we will exploit for local random walk teleportation. However, for the global DLA growth rule, particles originate from a distant source. We model this by taking the limit of the local harmonic measure as the starting point moves to infinity.

Definition 2.7 (Harmonic Measure from Infinity). *Let $K \subset \mathbb{Z}^2$ be a finite cluster and ∂K its adjacent empty boundary. The harmonic measure $H_{\partial K}(y)$ is the probability that a simple random walk starting from infinity first hits the boundary at the point $y \in \partial K$. It is defined as the limit:*

$$H_{\partial K}(y) = \lim_{|x| \rightarrow \infty} H_{\mathbb{Z}^2 \setminus (K \cup \partial K)}(x, y) = \lim_{|x| \rightarrow \infty} \mathbb{P}_x(S_\tau = y),$$

where $\tau = \inf\{n \geq 0 : S_n \in \partial K\}$ is the first hitting time of the boundary.

While the definition relies on an abstract limit from infinity, Lawler and Limic [5, Proposition 6.6.1] prove this limit strictly exists and defines a unique probability measure on ∂K . To calculate these probabilities locally in a computational setting, we evaluate the flow of random walkers into the cluster using the concept of discrete flux. Rather than simulating the entire path from infinity, the relative hitting probability at a specific boundary site can be determined strictly by the potential differences from its immediate free-space neighbours.

Definition 2.8 (Discrete Flux). *For a potential field $u : \mathbb{Z}^2 \rightarrow \mathbb{R}$ that vanishes on the boundary ∂K , the discrete flux into a boundary point $y \in \partial K$ is the sum of the potential differences from its free-space neighbours:*

$$\Phi(y) = \sum_{z \sim y, z \notin K \cup \partial K} (u(z) - u(y)) = \sum_{z \sim y, z \notin K \cup \partial K} u(z).$$

Proposition 2.9 (Harmonic Measure as Flux). *The harmonic measure at $y \in \partial K$ is exactly proportional to the discrete flux into y :*

$$H_{\partial K}(y) = \frac{\Phi(y)}{\sum_{w \in \partial K} \Phi(w)}.$$

Proof. See Lawler and Limic [5, Proposition 6.6.1], noting that the discrete Laplacian of the potential evaluated at the cluster boundary reduces exactly to our definition of the discrete flux. $\square$

To understand this link conceptually, we interpret the potential $u(x)$ as the density of a stream of random walkers originating from infinity. In the free space, $\mathcal{L}u(x) = 0$ dictates a conservation law: walkers entering a site equal those leaving. However, at a boundary site $y \in \partial K$, walkers arrive from neighbours but are absorbed, forming the cluster. The operator $\mathcal{L}u(y)$ represents this absorption rate, which is directly proportional to $\Phi(y)$.

This concludes the discrete formulation of the DLA process. The growth mechanism can be summarised in three rules:

1. The system is governed by a potential field $u(x)$ satisfying the discrete Laplace equation $\mathcal{L}u = 0$ in the exterior free space, with $u|_{\partial K} = 0$ and $u(x) \sim \ln|x|$ as $|x| \rightarrow \infty$.
2. The hitting probability at a boundary site y is dictated by the local flux $\Phi(y) = \sum_{z \sim y} u(z)$.
3. A new particle is added to ∂K strictly according to this harmonic measure $H_{\partial K}(y) \propto \Phi(y)$.

2.2 Continuous Potential Theory

We now provide an introduction to potential theory in the continuous setting. The fundamental shift is replacing the discrete random walk on a lattice with its continuous analogue: Brownian motion. Intuitively, Brownian motion can be viewed as the continuous scaling limit of a random walk.

Remark 2.10. The transition from the discrete simple random walk to continuous Brownian motion is formally guaranteed by Donsker's Invariance Principle [13, Theorem 5.22]. A full exposition of this functional limit theorem falls outside the scope of this text, but the intuitive link is enough for our purposes.

This section synthesises standard results regarding Brownian motion and potential theory, derived largely from Mörters and Peres [13], for the context of DLA. While fully rigorous proofs often require mathematics beyond the assumed knowledge of the intended audience, the fundamental theorems can be intuitively understood as the natural continuous extensions of the discrete framework established in the previous section.

2.2.1 Brownian Motion and the Laplacian

Since Brownian motion is the continuous engine of Off-Lattice DLA, we begin with its formal definition.

Definition 2.11 (Standard Brownian Motion). *A standard Brownian motion in $\mathbb{R}^d$ is a stochastic process $B = (B_t)_{t \geq 0}$ defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ taking values in $\mathbb{R}^d$, such that:*

1. $B_0 = 0$ almost surely.
2. The process has independent increments: for any $0 \leq s < t$, the increment $B_t - B_s$ is independent of the past history up to time s .
3. The increments are Gaussian: $B_t - B_s \sim \mathcal{N}(0, (t - s)I_d)$.
4. The paths $t \mapsto B_t$ are continuous almost surely.

In the study of 2D DLA, we will extensively utilise results from complex analysis. It is therefore natural to identify the plane $\mathbb{R}^2$ with the complex plane $\mathbb{C}$.

Definition 2.12 (Complex Brownian Motion). *A canonical complex Brownian motion Z_t is a process taking values in $\mathbb{C}$, defined as $Z_t = X_t + iY_t$, where X_t and Y_t are two independent, standard one-dimensional Brownian motions.*

In the discrete setting, we saw that harmonic functions ($\mathcal{L}f = 0$) evaluated along a random walk form a martingale. The continuous analogue relies on the continuous Laplace operator, $\Delta = \sum_{i=1}^d \frac{\partial^2}{\partial x_i^2}$. A twice continuously differentiable function $u : D \rightarrow \mathbb{R}$ defined on a domain $D \subset \mathbb{R}^d$ is called harmonic if $\Delta u(x) = 0$ for all $x \in D$.

Proposition 2.13 (Continuous Harmonic Functions are Martingales). *Let B_t be a standard d -dimensional Brownian motion and $D \subset \mathbb{R}^d$ be a domain. If $u : D \rightarrow \mathbb{R}$ is a harmonic function ($\Delta u = 0$), and τ_D is the first exit time from a bounded subdomain strictly contained in D , then the stopped process $M_t = u(B_{t \wedge \tau_D})$ is a martingale.*

Proof Sketch. We utilise the fact that a Gaussian probability density function $\mathbf{p}(t, x)$ inherently satisfies the heat equation:

$$\frac{\partial \mathbf{p}}{\partial t} = \frac{1}{2} \Delta \mathbf{p}.$$

Because standard Brownian motion is normally distributed, its probability distribution evolves over time according to this exact equation. When calculating the expected value of a function $u(B_t)$ over time, this relationship links the time derivative of the expectation directly to the spatial Laplacian of the function. Since u is harmonic, its Laplacian vanishes ($\Delta u = 0$). Consequently, the time derivative evaluates to zero, meaning the expected value is conserved. For a fully rigorous analytic proof, see Mörters and Peres [13, Theorem 2.51 and Corollary 2.53], while for an equivalent proof utilising Itô's formula (the stochastic analogue of the chain rule), see [13, Theorem 7.18]. $\square$

We additionally include an equivalent formulation of a harmonic function, which can be thought of as extending the idea of a harmonic function being the average of its neighbours to the continuous case.

Theorem 2.14 (Mean Value Property). *A locally integrable function $u : D \rightarrow \mathbb{R}$ is harmonic ($\Delta u = 0$) if and only if for every ball $B(x, r)$ strictly contained in the domain D :*

$$u(x) = \frac{1}{|\partial B(x, r)|} \int_{\partial B(x, r)} u(y) d\sigma(y),$$

where σ denotes the surface measure on the boundary sphere.

Proof. See Mörters and Peres [13, Theorem 3.2]. $\square$

2.2.2 The Continuous Dirichlet Problem

Just as in the discrete case, simulating DLA requires calculating the probability that a diffusing particle hits a specific segment of the cluster's boundary. Because we are now operating in a continuous domain $D \subset \mathbb{R}^d$, we can formalise these hitting probabilities using the harmonic measure.

Definition 2.15 (Continuous Harmonic Measure). *Let $D \subset \mathbb{R}^d$ be a bounded domain. The continuous harmonic measure $\omega_x = \omega_x(D, \cdot)$ is the probability measure on the boundary ∂D defined by:*

$$\omega_x(E) = \mathbb{P}_x(B_{\tau_D} \in E),$$

for Borel sets $E \subset \partial D$, where $\tau_D = \inf\{t > 0 : B_t \notin D\}$ is the first exit time.

Remark 2.16 (Continuous Harmonic Measure from Infinity). Similar to the discrete case, we define the harmonic measure from infinity by placing the starting point on a circle of radius R enclosing the cluster, and taking the weak limit of the hitting distribution as $R \rightarrow \infty$.

With this probability measure established, we can construct solutions to the continuous Dirichlet problem. Analytically, the Dirichlet problem seeks a function $u(x)$ that is harmonic within D and matches a specified continuous boundary condition $g(x)$ on ∂D . The probabilistic solution to this system mirrors the discrete Optional Stopping Theorem proof seen in Theorem 2.2.

Theorem 2.17 (Solution of the Continuous Dirichlet Problem). *Let $D \subset \mathbb{R}^d$ be a bounded domain and g be a continuous function on ∂D . The function defined by the expected value at the boundary:*

$$u(x) = \mathbb{E}_x[g(B_{\tau_D})] = \int_{\partial D} g(y) d\omega_x(y)$$

is harmonic in D and satisfies $\lim_{x \rightarrow y} u(x) = g(y)$ for regular boundary points $y \in \partial D$. A boundary point y is defined as regular for the complement of D if a Brownian particle starting at y immediately exits the domain almost surely, formally $\mathbb{P}_y(\tau_D = 0) = 1$.

Proof. See Mörters and Peres [13, Theorem 8.5(b)], although the fully rigorous proof using capacity theory is beyond our assumed scope. An elementary formulation is provided in [13, Theorem 3.12], but it relies on the Poincaré cone condition, a geometric requirement that every boundary point be accessible from the complement of D via a local cone. This is not necessarily satisfied by the deeply shielded fjords (channels between branches) in the idealised limit of a DLA cluster. $\square$

As in the discrete case, we consider the behaviour of the random process over infinite domains. As a consequence of the following lemma, 2D Brownian motion is neighbourhood recurrent, meaning it will revisit any local neighbourhood infinitely often.

Lemma 2.18 (Exit Probabilities from an Annulus). *Let B_t be a complex Brownian motion starting at z_0 within the annulus $A = \{z \in \mathbb{C} : r < |z| < R\}$. Let $T_A = \inf\{t > 0 : B_t \notin A\}$ be the first exit time. The probability that the process exits through the inner boundary $|z| = r$ is given by:*

$$\mathbb{P}_{z_0}(|B_{T_A}| = r) = \frac{\ln R - \ln |z_0|}{\ln R - \ln r}.$$

Taking the limit $R \rightarrow \infty$ yields $\mathbb{P}(|B_{T_A}| = r) \rightarrow 1$, proving planar Brownian motion is neighbourhood recurrent.

Proof. See Mörters and Peres [13, Theorem 3.18]. $\square$

2.2.3 Conformal Invariance

In the discrete setting, the recurrence of the 2D walk forced us to abandon the standard Green's function and construct a regularised potential kernel to solve for hitting probabilities on unbounded domains. In the continuous setting, rather than taking a parallel approach to the discrete section, we instead exploit a fundamental property of planar Brownian motion: conformal invariance. While this theoretically provides a complete solution to the Dirichlet problem on the full, complex cluster boundary [14], in practice, we utilise this property as the mathematical backbone for the optimisations employed in our Off-Lattice and Hybrid algorithms.

Theorem 2.19 (Conformal Invariance of Brownian Motion). *Let $U \subset \mathbb{C}$ be a domain and $f : U \rightarrow V$ be a non-constant analytic function. If Z_t is a complex Brownian motion in U , and τ_U is its first exit time from U , then the image process $W_t = f(Z_t)$ is, up to a random time change, a complex Brownian motion in V .*

Proof. See Mörters and Peres [13, Theorem 7.20]. $\square$

Because DLA is governed by the spatial distribution of hitting points, the time change introduced by f is irrelevant to our model, leading directly to the invariance of the hitting probabilities.

Corollary 2.20 (Conformal Invariance of Harmonic Measure). *Let $f : D \rightarrow D'$ be a conformal map. The harmonic measure is invariant under f :*

$$\omega_z(D, E) = \omega_{f(z)}(D', f(E))$$

for any Borel set $E \subset \partial D$.

Proof. See Mörters and Peres [13, Theorem 7.23(a)] □

2.2.4 Solvable Geometries

To utilise conformal invariance, we need to know the explicit probability density function for the hitting probabilities on our reference domains.

Definition 2.21 (Poisson Kernel). *Let σ be the surface measure on ∂D . If the harmonic measure is absolutely continuous with respect to σ , the Poisson kernel $P_D(z, \zeta)$ is defined as the Radon-Nikodym derivative:*

$$P_D(z, \zeta) = \frac{d\omega_z}{d\sigma}(\zeta).$$

It represents the probability density of the Brownian motion exiting D at $\zeta \in \partial D$ given it started at $z \in D$.

We can now express the conformal invariance of the harmonic measure in terms of the Poisson kernel. By definition, the harmonic measure of a Borel set $E \subset \partial D$ is the integral of the Poisson kernel over E .

Applying the conformal invariance corollary to a map $f : D \rightarrow \mathbb{D}$ to the unit disk such that $f(z) = 0$, and noting that the Poisson kernel on the disk from the origin is uniform ($P_{\mathbb{D}}(0, w) = \frac{1}{2\pi}$), we have:

$$\int_E P_D(z, \zeta) d\sigma(\zeta) = \omega_0(\mathbb{D}, f(E)) = \int_{f(E)} \frac{1}{2\pi} d\sigma(w).$$

Under the change of variables $w = f(\zeta)$, the boundary arc length element transforms as $d\sigma(w) = |f'(\zeta)| d\sigma(\zeta)$. Substituting this into the right-hand integral yields:

$$\int_E P_D(z, \zeta) d\sigma(\zeta) = \int_E \frac{1}{2\pi} |f'(\zeta)| d\sigma(\zeta).$$

Since this equality holds for any Borel set E , the integrands must be strictly equal, yielding the fundamental relationship:

$$P_D(z, \zeta) = \frac{1}{2\pi} |f'(\zeta)|. \tag{2.2}$$

We now apply (2.2) to derive the exact solutions for two fundamental domains: the upper half-plane ($\mathbb{H}$) and the unit disk ($\mathbb{D}$).

Theorem 2.22 (Poisson Kernel for $\mathbb{H}$). *Consider the upper half-plane $\mathbb{H} = \{z \in \mathbb{C} : \text{Im}(z) > 0\}$. The probability density of a Brownian particle starting at $z = x + iy$ (with $y > 0$) hitting the real line at $u \in \mathbb{R}$ is given by the Cauchy distribution:*

$$P_{\mathbb{H}}(z, u) = \frac{1}{\pi} \frac{y}{(u - x)^2 + y^2}.$$

Proof. We construct a conformal map $\phi : \mathbb{H} \rightarrow \mathbb{D}$ that sends the starting point z to the origin 0, and the real line $\mathbb{R}$ to the boundary of the unit disk, utilising the Möbius transformation:

$$\phi(w) = \frac{w - z}{w - \bar{z}}.$$

We compute the derivative with respect to w :

$$\phi'(w) = \frac{z - \bar{z}}{(w - \bar{z})^2}.$$

Evaluating this derivative at a boundary point $u \in \mathbb{R}$:

$$|\phi'(u)| = \frac{|2iy|}{|u - \bar{z}|^2} = \frac{2y}{(u - x)^2 + y^2},$$

where we have used the fact that $y > 0$. Thus, applying (2.2), the Poisson kernel is $\frac{1}{2\pi}|\phi'(u)| = \frac{1}{\pi} \frac{y}{(u-x)^2 + y^2}$. $\square$

Theorem 2.23 (Poisson Kernel for $\mathbb{D}$). *The Poisson kernel for the unit disk $\mathbb{D}$, representing the exit density at angle θ for a particle starting at $z \in \mathbb{D}$, is given by:*

$$P_{\mathbb{D}}(z, e^{i\theta}) = \frac{1}{2\pi} \frac{1 - |z|^2}{|e^{i\theta} - z|^2}.$$

Proof. We seek an automorphism of the disk $\psi : \mathbb{D} \rightarrow \mathbb{D}$ that maps the starting point z to the origin 0. This is given by the Möbius transformation: $\psi(w) = \frac{w - z}{1 - \bar{z}w}$. The derivative at the boundary point $w = e^{i\theta}$ is:

$$\psi'(e^{i\theta}) = \frac{1 - |z|^2}{(1 - \bar{z}e^{i\theta})^2}.$$

Since $|1 - \bar{z}e^{i\theta}| = |e^{-i\theta} - \bar{z}| = |\overline{e^{i\theta} - z}| = |e^{i\theta} - z|$, the magnitude of the derivative gives $|\psi'(e^{i\theta})| = \frac{1 - |z|^2}{|e^{i\theta} - z|^2}$. Therefore, the Poisson kernel is exactly $\frac{1}{2\pi}|\psi'(e^{i\theta})| = \frac{1}{2\pi} \frac{1 - |z|^2}{|e^{i\theta} - z|^2}$. $\square$

Corollary 2.24 (Poisson Kernel for the Exterior of $\mathbb{D}$). *Let $D_{ext} = \{z \in \mathbb{C} : |z| > 1\}$. The harmonic measure of the unit circle relative to a point $z \in D_{ext}$ is given by the density:*

$$P_{ext}(z, e^{i\theta}) = \frac{1}{2\pi} \frac{|z|^2 - 1}{|z - e^{i\theta}|^2}.$$

Proof. Consider the geometric inversion $a = 1/\bar{z}$, which maps $D_{ext} \rightarrow \mathbb{D}$. By the conformal invariance of harmonic measure, the exit distribution for z is identical to that of a . Substituting a into the interior kernel (Theorem 2.23) and simplifying yields the result. $\square$

This concludes the continuous formulation of the potential theory underlying DLA. The framework parallels the discrete setting, but replaces lattice regularisation with conformal geometry:

1. The system is governed by a continuous potential field $u(x)$ satisfying the continuous Laplace equation $\Delta u = 0$.
2. Hitting probabilities are framed as continuous Dirichlet problems and solved via the probabilistic formulation in Theorem 2.17.
3. Rather than solving these problems over complex boundaries, conformal invariance allows us to map difficult domains to simple, solvable geometries like $\mathbb{H}$ and $\mathbb{D}$. The exact exit distributions are given by their respective Poisson kernels, which serve as the continuous analogue to the discrete flux.

Chapter 3

Algorithms

3.1 Introduction

Simulating DLA presents a significant computational challenge. Directly simulating the random walk of each particle from a large distance is prohibitively expensive. As the mean squared displacement of a diffusing particle grows linearly with time, the number of steps required to traverse a region of radius R scales as $\mathcal{O}(R^2)$. To generate large enough clusters for useful analysis, we employ accelerated algorithms that exploit the theory developed in Chapter 2.

In this chapter, we develop three distinct models:

1. **On-Lattice DLA:** A discrete implementation adapted from the algorithm by Loh [15]. Space is restricted to a regular grid, and a "particle" is merely the coordinate index of an active cell.
2. **Off-Lattice DLA:** A continuous implementation using conformal mappings adapted from Bell [16]. Here, particles are treated as physical disks in $\mathbb{R}^2$ undergoing Brownian motion.
3. **Hybrid DLA:** A novel model that combines the continuous spatial diffusion of the Off-Lattice approach with the discrete aggregation mechanics of the On-Lattice model.

These models all share roughly the same structure, which we split into four stages: Initialisation, Far-Field Diffusion, Near-Field Diffusion and Aggregation. We will detail each of the stages for the first two models in the following sections. We will then present the Hybrid model in Section 3.6, which we implement as a variation of the Off-Lattice model.

3.2 Initialisation

The mathematical definition of DLA requires particles to originate at infinity. In practice, we do this by launching particles from a boundary $\partial\Omega$ sufficiently far from the cluster K . We now detail how this is achieved in both the Off-Lattice and Lattice models.

3.2.1 Off-Lattice: The Launching Circle

In the continuous setting, initialising at infinity is equivalent to initialising uniformly on a bounding circle. To see this, let R_{max} be the radius of the smallest circle centred at the origin containing the cluster. We define a launching circle of radius $R_l > R_{max}$.

Consider the conformal map $\phi(z) = \frac{R_l}{z}$. This creates a bijection between the exterior of the launching circle and the interior of the unit disk $\mathbb{D}$, mapping infinity directly to the origin. As established by evaluating the disk's Poisson kernel at the origin (Theorem 2.23 with $z = 0$), the harmonic measure is uniform. Consequently, the hitting distribution of a Brownian motion starting at infinity on the circle $|z| = R_l$ is also uniform. Thus, our Off-Lattice algorithm simply samples a starting angle $\theta \sim U[0, 2\pi)$ and initialises the walker at $z_0 = R_l e^{i\theta}$.

Note also that provided the circle contains the entire cluster there is no size restriction, so we can scale it dynamically, thereby improving the efficiency of the simulation.

3.2.2 On-Lattice: The Fuzzy Annulus

However in the discrete setting, the above approach introduces a bias. As detailed in [15], projecting a circle onto a discrete grid forces the starting positions to snap to the nearest integer coordinates. This creates a ‘lattice aliasing’ effect (also referred to as a Moiré interference pattern). For large enough clusters, this bias is sufficient to distort key metrics including the fractal dimension.

To eliminate this, we implement Loh’s Fuzzy Annulus method. Instead of a fixed radius R , the starting position is sampled from a probability distribution spread over an annulus. This probability distribution utilises the Kaiser-Bessel function I_0 , which originates from signal processing [17]:

$$P(x) \propto \frac{I_0(x)}{I_0(R_l)}.$$

By sampling from this distribution, we smooth out the grid aliasing effects, reducing the bias to computationally negligible levels.

Like in the Off-Lattice case, we can scale the launching annulus dynamically, but to further reduce the bias, the inner boundary of the annulus is at $2R_{max}$.

Remark 3.1 (Rejection Sampling). Rather than directly sampling from this distribution, we use an optimisation technique called ‘rejection sampling’, which we explain in Appendix B.3.

3.3 Far-Field Diffusion

Once initialised, or indeed at any point during the diffusion, the walker may be far from the cluster. To improve the efficiency of our algorithms, we utilise results from Chapter 2 to quickly return the walker to near the cluster.

3.3.1 Off-Lattice: Re-injection from Infinity

In the continuous case, if the walker strays outside the launching circle ($|z_{out}| > R_l$), we exploit the conformal invariance established in Section 2.2 to immediately map it back to this boundary.

We do this by sampling the first hitting point on the circle directly. As shown in Corollary 2.24, the return probability is governed by the exterior Poisson kernel. To efficiently sample from this distribution, we first note that the conformal invariance of Brownian motion establishes a statistical equivalence between geometries: to sample the hitting distribution on a physical boundary $\partial\Omega$, we can sample the hitting distribution on a reference domain ∂D (where the distribution is known, e.g., Uniform) and transform the result via the conformal map $f : D \rightarrow \Omega$. This is the key idea for applying conformal maps to our DLA algorithm [16].

In this instance, we map the uniform exit distribution on the unit circle to the exterior harmonic measure via the following steps:

1. Normalisation: We normalise the walker’s position z to $r = |z|/R_l$ and map it to the interior point $a = 1/r$ via inversion. Note that r and a are real numbers.
2. Möbius Transformation: We generate a uniform variate ζ on the unit circle (representing an isotropic exit from the origin). We then apply the real-axis Möbius map:

$$w = \frac{r\zeta + 1}{r + \zeta}.$$

This maps the unit circle to itself, sending the origin to the point a inside the disk. By the conformal invariance of harmonic measure, the transformed variate w is now distributed according to the harmonic measure relative to a .

3. Reconstruction: Inverting now does not change the distribution on the unit circle, so w is distributed exactly according to the exterior harmonic measure. We rotate w by $\arg(z_{out})$ and scale by R_l to obtain the true re-entry position.

This effectively folds the infinite exterior space back onto the launch circle in $\mathcal{O}(1)$ operations.

Remark 3.2. This ‘no-killing’ approach was formally introduced by Menshutin and Schur [18] as a ‘killing-free’ algorithm, where they sampled from the exact hitting distribution (via the Poisson Kernel). We instead adopt Bell’s conformal mapping approach, although we map to the unit disk while Bell maps to a straight line. We do this to reduce computational overhead, as it replaces complex Cauchy sampling with trivially fast uniform angular sampling.

3.3.2 Lattice: Walk-to-Line

In the discrete setting, we cannot simply use a conformal map to transport a walker from infinity to the cluster’s boundary, as the grid lacks the continuous symmetries required for conformal invariance. As an alternative, we employ Loh’s Walk-to-Line acceleration method.

When a walker wanders outside the cluster’s bounding box, we project its position onto the nearest face of the box. We treat this face as an infinite absorbing line $\partial\Lambda = \{(x, 0) : x \in \mathbb{Z}\}$ and the walker’s position as $z_0 = (0, h)$ in the upper half-plane $y > 0$. The problem reduces to sampling the exit probability on the line, the discrete analogue to Theorem 2.22.

3.3.2.1 Discrete Method of Images

As established in Chapter 2, the exit probability (harmonic measure) at a boundary point $(x, 0)$ is proportional to the discrete flux into that point. Since the potential $u(z)$ vanishes on the boundary, and $(x, 1)$ is the only neighbour residing in the free-space domain $y > 0$, the flux is simply the value of the potential at the immediate free-space neighbour:

$$P_{\text{lattice}}(x) \propto \Phi(x, 0) = u(x, 1).$$

To calculate this, we construct a potential $u(z)$ that is harmonic in the upper half-plane but strictly satisfies the Dirichlet boundary condition $u(x, 0) = 0$ (representing the absorbing line). We achieve this via the Method of Images.

The key idea is to place a virtual negative walker (or sink) at the reflected position $(0, -h)$, opposite the true walker at $(0, h)$. Due to the symmetry of the lattice, the potential generated by the real walker at any point on the boundary line $y = 0$ is identical to the potential generated by the image. By subtracting the image potential from the source potential, these contributions cancel out exactly along the line, ensuring $u(x, 0) = 0$ by construction.

Using the discrete potential kernel $a(z)$ defined in Definition 2.3, the resulting field is:

$$u(x, y) = a(x, y + h) - a(x, y - h).$$

Substituting this into the flux equation ($y = 1$), we obtain the exact unnormalised probability of the walker hitting the line at relative coordinate x :

$$P_{\text{lattice}}(x) = a(x, 1 + h) - a(x, 1 - h).$$

Remark 3.3. To implement this, we follow Loh exactly. For $h \leq 60$, we evaluate this using his pre-calculated lookup table of $a(z)$. For larger distances, we use the asymptotic expansion of the potential kernel, given in the appendix of [15].

Remark 3.4. Computing the normalisation constant for $P_{\text{lattice}}(x)$ over the infinite line is computationally problematic. Instead, we once again employ rejection sampling (details in Appendix B.3).

3.3.2.2 The Killing Circle

While the Walk-to-Line method is theoretically valid for any distance, practical implementations are bound by the limits of 64-bit integer arithmetic. If a walker wanders exceptionally far from the cluster, the coordinate calculations will overflow. To prevent numerical instability, we define a hard “Killing Radius” at $R_{\text{kill}} = 10^{14}$. In the extremely rare event that a walker’s random step projects it beyond this limit, the particle is discarded and a new one is initialised at the launching annulus.

Although 2D random walks are recurrent, a particle returning from such an extreme distance effectively loses all angular ‘memory’ of its escape trajectory. The deviation between its true return distribution and a fresh uniform sample from infinity scales proportionally to the ratio of the cluster radius to the killing radius, R_B/R_{kill} [15]. Because our maximum expected cluster radius is $R_B \approx 10^4$, replacing the errant walker introduces a bounded bias of only $\sim 10^{-10}$, ensuring numerical stability without compromising the harmonic measure.

3.4 Near-Field Diffusion

As the walker approaches the cluster, the step size must decrease to avoid overshooting the boundary.

3.4.1 Off-Lattice: Walk-to-Circle

At each step, we calculate the shortest Euclidean distance to the cluster, $d_{\min}$. Details of this calculation are included in Appendix B.2. We can construct a circle C of radius $r = d_{\min} - \epsilon$ (where ϵ is a small tolerance, set to 2 in our simulation). Since the interior of C is empty, the potential is harmonic. By the isotropy of Brownian motion (Theorem 2.23 with $z = 0$), the exit distribution is uniform. We therefore advance the walker (at position z) to a random point on the boundary of C :

$$z_{\text{new}} = z + re^{i\theta}, \quad \theta \sim U[0, 2\pi).$$

This allows the walker to adaptively skip over large empty regions, taking large steps in the void and small steps in the fjords.

3.4.2 Lattice: Walk-to-Square

In the discrete setting, circles do not map cleanly to the grid. Instead, we employ Loh’s Walk-to-Square method. This relies on the pre-calculation of exit probabilities for a hierarchy of empty square sizes $L_k = 2^k$ (e.g., from 4×4 up to 512×512).

Unlike a circle, the exit distribution on the boundary of a square is not uniform, as lattice points near the mid-edges are hit more frequently than the corners. We determine these exact probabilities by solving the discrete Dirichlet problem $\mathcal{L}u = 0$ (Theorem 2.2) for each square size S_k . For simplicity we used Loh’s precomputed solutions (after verification).

Remark 3.5 (Alias Sampling). To sample from these non-uniform distributions efficiently, we calculate them offline and store them using Walker’s Alias Method (detailed in Appendix B.4). This allows us to sample the discrete exit location in $\mathcal{O}(1)$ time, regardless of the square size.

The simulation step proceeds as follows:

1. Determine the largest square S_k centred at the walker that does not intersect the cluster.
2. Query the pre-computed Alias Table for size k to generate a boundary coordinate z_{exit} .

3. Update the walker's position to z_{exit} .

This allows the walker to leap across large voids while preserving the exact non-uniform statistics of the lattice random walk.

3.5 Aggregation

The final stage is the termination of the walk.

3.5.1 Lattice: Sticky Boundary

The aggregation condition on $\mathbb{Z}^2$ is straightforward. We define the boundary ∂K as the set of all empty sites adjacent to occupied sites, as we did in chapter Chapter 2. The walk terminates when the particle steps onto a site $x \in \partial K$. The site x is then marked as occupied ($x \in K$), and the boundary ∂K is updated locally.

3.5.2 Off-Lattice: Exact Adhering

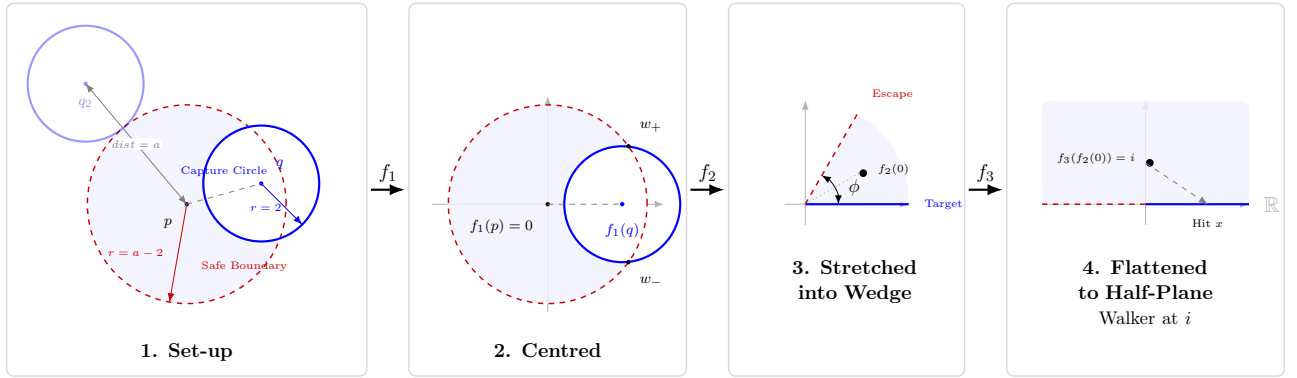


Figure 3.1: The conformal mapping sequence for exact collision. The physical domain (1), formed by the intersection of the safe region and the target exterior, is centred (2), mapped to an infinite wedge (3), and finally opened to the upper half-plane (4) where the exit position is sampled. The shaded blue region in each represents the domain.

In the continuous model, aggregation occurs when the walker touches a cluster particle (i.e., the center-to-center distance is 2, since the radius of each particle is 1). Simulating this directly with fixed time-steps is not ideal, due to the potential errors of the walker grazing the cluster or overshooting.

To resolve this, we adopt the analytic method proposed in [16]. When the walker is at position p , close to a target particle centred at q but constrained by a second-nearest neighbour q_2 at distance a , we define a local domain Ω . As illustrated in Figure 3.1 (Stage 1), Ω is the lune (crescent-shape) formed by the intersection of the safe escape circle (radius $a - 2$, centred at p) and the exterior of the capture circle (radius 2, centred at q).

To sample the exact exit position from Ω without time-stepping errors, we conformally map this lune to the upper half-plane $\mathbb{H}$ in three stages:

1. **Centring (f_1):** We first translate and rotate the system so that the walker p lies at the origin and the target q lies on the positive real axis:

$$f_1(z) = |q - p| \frac{z - p}{q - p}.$$

As shown in Figure 3.1 (Stage 2), we identify the intersection points of the two circles in this transformed space, denoted w_+ and w_- .

2. **Stretching (f_2):** We apply a Möbius transformation that maps the intersection point w_+ to the origin and w_- to infinity:

$$f_2(z) = \frac{z - f_1(w_+)}{z - f_1(w_-)}.$$

This transformation stretches the circular boundaries into two straight rays emanating from the origin, forming an infinite wedge (Figure 3.1, Stage 3). The angle ϕ between these rays is determined by the intersection geometry. The walker now lies somewhere on the segment of the unit circle within this wedge.

3. **Flattening (f_3):** Finally, we open the wedge into the full upper half-plane using a power map:

$$f_3(z) = z^{\pi/\phi}.$$

This maps the target boundary to the positive real line and the escape boundary to the negative real line (Figure 3.1, Stage 4). Due to the symmetry of the original lune about the $p - q$ axis, the walker is mapped to the point i .

In this final domain, the problem reduces to a standard Brownian motion starting at i and hitting the real line. We sample the hitting point from the Cauchy distribution (Theorem 2.22 evaluated at $z = i$, which easily integrates to give the CDF) and map it back to the physical domain using the inverse composition $f_1^{-1} \circ f_2^{-1} \circ f_3^{-1}$. If the mapped point lies on the target circle, the particle adheres; if it lies on the safe circle, the walker is effectively reset, and diffusion continues.

3.6 The Hybrid Model

To disentangle the source of anisotropy in DLA, we introduce a Hybrid model.

3.6.1 Motivation and Design

We hypothesise that the characteristic cross-shape of Lattice DLA arises primarily from the restriction of particles to fixed grid sites, rather than the discrete transport dynamics of the lattice walk. To rigorously test this, we require a strictly controlled Hybrid model where the transport is purely isotropic (continuous Brownian motion), but the final aggregation is constrained to a discrete grid geometry.

The critical design choice is the microscopic aggregation mechanism. We primarily evaluated two geometric strategies:

1. **The Sticky Boundary:** A walker aggregates if it enters any square cell within a predefined boundary. We rejected this because the square capture zone imposes lattice anisotropy onto the incoming probability flux before aggregation occurs, preventing a clean separation of transport and packing effects.
2. **The Twist Method (Chosen):** The walker behaves exactly as a hard sphere in continuous space. Upon exact analytical collision with the cluster, the particle's continuous angle of impact is projected to the nearest valid lattice site, and the particle is snapped into place. By isolating the discrete packing to the exact instant of impact, the arrival statistics are entirely dependent on the continuous harmonic measure.

We note a conceptual similarity to the ‘antenna method’ explored by Menshutin and Shchur [11]. However, their study postulated that shifting a continuous walker to a predefined axial antenna upon contact is functionally identical to a pure square-lattice model, without providing direct comparative evidence. Consequently, our subsequent quantitative comparison of these transport models provides a novel empirical contribution. Furthermore, strict algorithmic equivalence cannot be assumed, as the exact conformal adhesion technique utilised in our model had not yet been developed at the time of their publication.

3.6.2 Implementation

The algorithm proceeds identically to the Off-Lattice simulation until the collision condition $|z_{walker} - z_{cluster}| \leq 2$ is met (since particle radius $r_p = 1$).

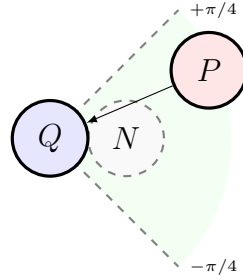


Figure 3.2: The angular window for hybrid aggregation. If a walker P collides with particle Q within the angle $|\phi| < \pi/4$ (shaded region), it is snapped to the nearest lattice slot N . We can assume N is empty because if it were occupied, its own collision shell would have intercepted the walker before it could reach Q within this specific angle.

When a collision occurs, the position of the walker relative to the target particle is $\Delta z = \Delta x + i\Delta y$. We then determine the nearest cardinal lattice direction simply by comparing the absolute Cartesian components. For instance, the primary candidate is defined as the horizontal neighbour if $|\Delta x| > |\Delta y|$. The particle is then snapped to this adjacent lattice coordinate.

A potential concern is whether the snapped lattice site is available. We provide a simple geometric argument to justify that the target slot is always empty.

Consider a walker P hitting a cluster particle Q at angle ϕ . The snapped destination is the lattice neighbour N closest to the point of impact. For a grid of spacing 2, the neighbour N is located at distance 2 from Q (centre-to-centre). If N were already occupied, its own collision shell (radius 2) would overlap the collision shell of P .

Simple trigonometry shows that for any impact angle ϕ closer to N than any other neighbour (i.e., $|\phi| < \pi/4$), the distance from the impact point to N is strictly less than 2:

$$|z_{hit} - z_N| = |2e^{i\phi} - 2| = 2\sqrt{2 - 2\cos\phi}.$$

For $|\phi| < \pi/4$, this distance is at most $2\sqrt{2 - \sqrt{2}} \approx 1.53$, which is well inside the collision radius of 2. This implies that if N were occupied, the walker would have intersected N ’s collision shell before reaching P ’s surface. Since the algorithm halts at the first contact, the target slot N must be empty. This is visualised in Figure 3.2.

Remark 3.6 (Numerical Stability). While the geometric proof analytically guarantees that the primary candidate slot is empty, practical implementations are subject to microscopic floating-point rounding errors. To ensure algorithmic robustness against this numerical noise, we include a zero-cost fallback mechanism: if the primary slot erroneously registers as occupied, the walker is safely routed to the secondary cardinal direction.

Chapter 4

Simulation Results

The complete implementation of the algorithms described in Chapter 3, along with the rest of the codebase for this project, is available at: https://github.com/anonmathstudent/partb_dla. We implemented a Python/Numba framework combining the accelerated lattice, off-lattice, and hybrid DLA algorithms studied in this dissertation.

Using this code, we generated 1,000 clusters of size $N = 10^6$ particles and 100 clusters of size $N = 10^7$ particles for each of the three models (Off-Lattice, Lattice, Hybrid), each with an independent seed. We then stored the results as standardised 2D coordinate arrays in compressed NumPy files (`.npz` format).

In the subsequent chapter, we will perform a more rigorous analysis, designed to expand the results of [9] and compare across the three models. Before that, we first visualise the clusters to gain an intuitive understanding of the structural similarities and differences.

Figure 4.1 and Figure 4.2 present a visual comparison of the clusters generated by the three models, with the former providing a visualisation of some randomly chosen clusters of size $N = 10^7$, and the latter showing the spatial density maps of the entire ensemble of clusters at both $N = 10^6$ and $N = 10^7$ scales.

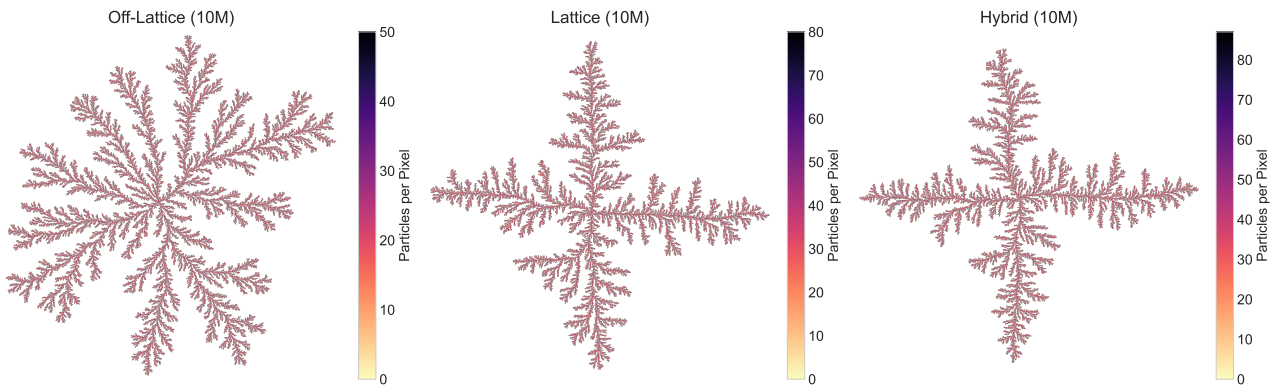


Figure 4.1: Coarse-grained spatial density maps for representative 10^7 -particle clusters across the three simulation models. To visualise the localised mass concentration, the spatial bounding box of each cluster is discretised into a high-resolution 2D grid ($2^{11} \times 2^{11}$ pixels). A spatial histogram is computed by counting the exact number of particles occupying each discrete pixel bin, with empty bins masked for clarity. A linear colour scale is applied to the resulting particle counts, where darker regions indicate an proportionately higher local mass density.

There are some immediate observations that can be made from these visualisations, with the key one being the similar cross-like structure between the Lattice and Hybrid models, while the Off-Lattice model appears much more uniform, resulting in near circular density maps in Figure 4.2. The remainder of this thesis will focus largely on expanding upon this observation.

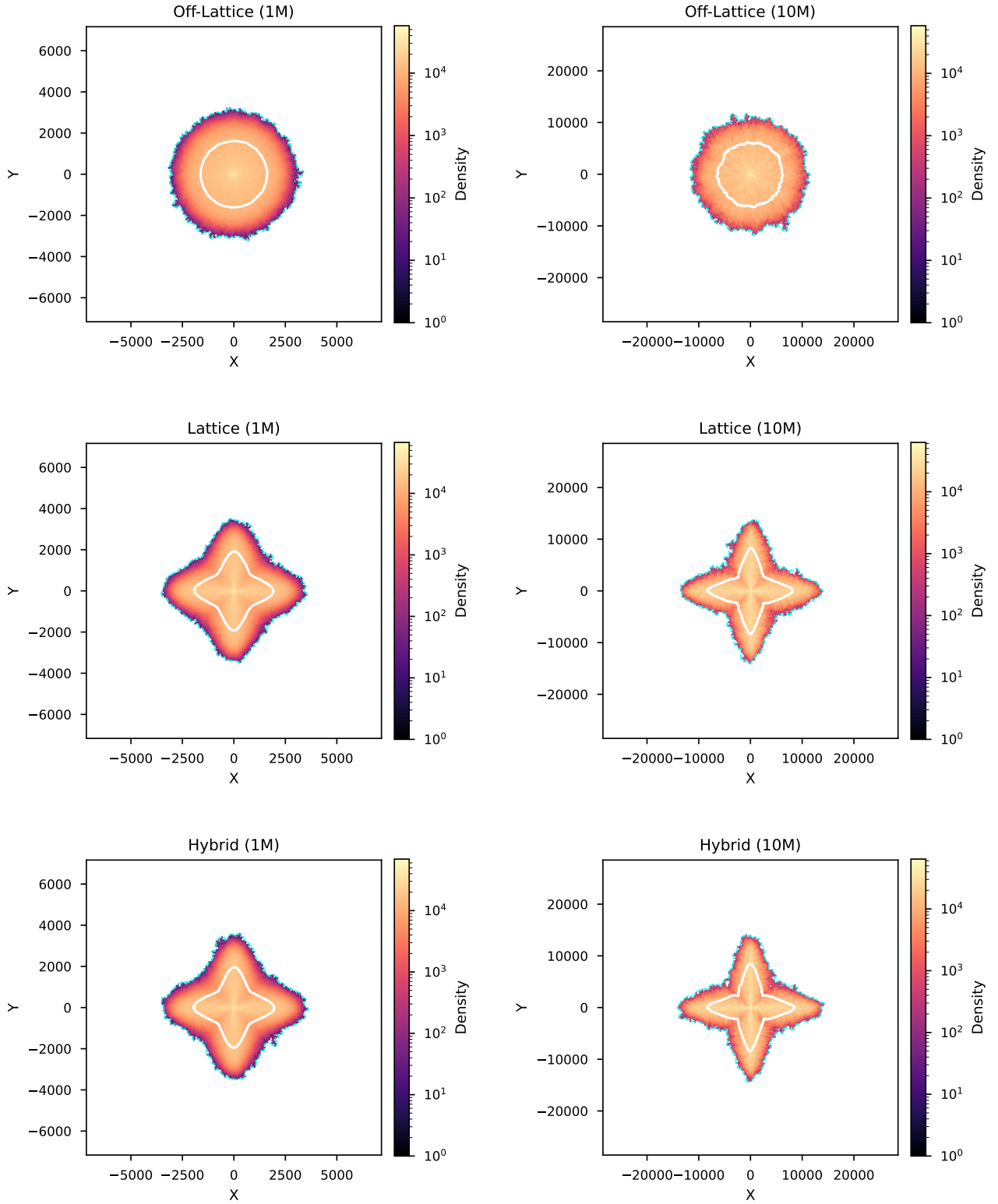


Figure 4.2: Spatial density maps of the simulated models at 10^6 and 10^7 scales. Unlike the single-cluster heatmaps, these visualisations aggregate the entire ensemble of simulated clusters into a unified 2D spatial histogram. The logarithmic colour intensity represents the cumulative frequency of particles occupying each spatial bin. Overlay contours denote the maximum absolute radial extent (dashed cyan) and the average radial mass distribution (solid white), also known as the angular radius of gyration R_g .

Chapter 5

Fractal Dimension of DLA

5.1 Introduction to Dimension

While DLA has been studied for over forty years, finding an exact mathematical proof for the fractal dimension of standard 2D radial growth remains an unsolved problem, beyond the rigorous lower bound of $D \geq 1.5$ established by Kesten [7]. Exact results currently exist only for highly constrained geometries, such as Off-Lattice DLA grown on an infinite line [19]. There are also multiple theoretical predictions of a dimension of $D = 5/3 \approx 1.667$, derived using various approaches including mean-field theory [20] and analytic diamond-shaped boundary models [21]. However, these approaches rely on simplifying assumptions, such as coarse-graining the local density fluctuations. Consequently, much of the literature relies upon large-scale computational approximations, though as we shall outline below, there are subtleties in how these dimensional metrics are defined and interpreted.

Before analysing our results, we must first establish what is meant by ‘dimension’. Dimension is a property that describes how an object fills space as it scales. For standard Euclidean geometries, this scaling relationship is an integer. For instance, scaling a solid 2D shape by a factor of 2 increases its area by $2^2 = 4$. However, for fractal objects, this scaling relationship does not resolve to an integer; scaling the fractal up by a factor of 2 increases its mass by 2^D , where D is a non-integer known as the fractal dimension.

For an in-depth explanation of how we formalise this notion, see [22]. Here, we provide a brief motivation and definition of the two metrics relevant for our purposes: box-counting dimension and growth rate.

5.1.1 Box-Counting Dimension (D_{box})

We first define the box-counting (or Minkowski) dimension. We motivate the definition by considering the power law of scaling.

Consider covering a standard Euclidean object with a uniform grid of boxes of side length ϵ . If the object is a 1D line of length L , the number of boxes required is $N(\epsilon) \propto L/\epsilon \sim \epsilon^{-1}$. If the object is a 2D square of area A , the number of boxes is $N(\epsilon) \propto A/\epsilon^2 \sim \epsilon^{-2}$. This fits our intuition for dimension for simple objects. We generalise this scaling relationship for an object of arbitrary dimension D_{box} as:

$$N_\epsilon(F) \sim \epsilon^{-D_{box}}$$

To isolate the dimension D_{box} , we take the logarithm of both sides and rearrange for D_{box} , giving:

$$D_{box} \approx \frac{\log N_\epsilon(F) - c}{-\log \epsilon}$$

where c is a constant of proportionality. As $(\epsilon \rightarrow 0)$, c becomes negligible, resulting in the following definition:

Definition 5.1 (Box-Counting Dimension). *Let $N_\epsilon(F)$ be the smallest number of ϵ -mesh cubes (boxes of side length ϵ) that intersect a set F . The box-counting dimension D_{box} is defined as:*

$$D_{box} = \lim_{\epsilon \rightarrow 0} \frac{\log N_\epsilon(F)}{-\log \epsilon}$$

assuming this limit exists.

Practically, by measuring the number of occupied boxes $N_\epsilon(F)$ for various box sizes ϵ and plotting them on a log-log graph, D_{box} is extracted directly from the negative slope of the regression line.

5.1.2 Growth Rate

Much of the literature on DLA uses a different metric for dimension, related to the growth rate of the cluster. The widely reported fractal dimension of $D \approx 1.71$ (e.g. [23] and [18]) is a result of this definition, which we now introduce, before discussing its relationship to the box-counting dimension.

We first follow [9] in considering a more robust metric than the maximal radius to represent the spatial extent of a cluster: the radius of gyration, R_g , using their definition.

Definition 5.2 (Radius of Gyration). *For a discrete aggregate of N identical particles located at positions $\mathbf{r}_i$, the radius of gyration, R_g , is the root-mean-square distance of the particles from the origin (seed particle), defined as:*

$$R_g = \sqrt{\frac{1}{N} \sum_{i=1}^N |\mathbf{r}_i|^2}$$

Similar to the box-counting dimension, there is a power law relationship between the radius of gyration and the number of particles in the cluster:

$$R_g \sim N^\beta$$

where β is defined as the growth rate exponent.

Compare this now to the power law satisfied by the box counting dimension for a cluster of N particles $R \sim N^{1/D_{box}}$. Equating them suggests another definition of dimension based on the growth rate, given by $D_\beta = \frac{1}{\beta}$.

This is the metric that is commonly used in the DLA literature to represent the fractal dimension of the cluster, and is often assumed to be equivalent to the box-counting dimension. However, this is not always the case, as the following simple example from [9] illustrates:

Consider an aggregate composed of $2t$ particles. Half of these particles (t) form a dense 2D disk of radius $R_{disk} \propto \sqrt{t}$. The other half (t) form a 1D interval (a line) of length $L \propto t$. The box dimension D_{box} is determined by the solid 2D disk, as it is the densest component, hence $D_{box} = 2$. However, the radius scales as $R \propto t$, and since the total mass is $N = 2t \propto t$, the radius scales linearly with the mass ($R \propto N^1$). This means that the growth rate is $\beta = 1$, yielding a Growth Rate Dimension of $D_\beta = 1$.

Therefore we may need to be more careful in interchanging between these two metrics, and we will analyse how they vary and evolve across the different models, expanding on the results from [9] specifically for the Lattice case.

5.2 Methodology

To compute both D_{box} and D_β , we rely on performing linear regression on log-log relationships. How this line is fit, and the potential error here, is not widely covered in the literature, which largely uses standard Ordinary Least Squares (L^2) regression.

We provide a novel comparison of the standard L^2 regression with a more robust alternative, the Theil-Sen estimator. The Theil-Sen estimator calculates the median of all pairwise slopes between data points, making it resistant to outliers and boundary effects that can skew L^2

regression. As well as providing insight into the effect of the choice of regression method on the resulting dimension (for both dimension metrics), it also served to identify the valid scaling regime for D_{box} .

5.2.1 Box Sizes for Box-Counting Dimension

When we fit the grids with box sizes comparable to the particle size (i.e., $\epsilon < 16$ pixels), the box-count is dominated by the local 1D topology of individual branches, and eventually saturates at the trivial 0D limit of counting individual particles. Similarly, as the boxes get large ($\epsilon > R/16$ pixels, where R is the cluster radius), the count is dominated by finite-size effects, and eventually saturates at the 0D limit of counting the entire cluster as one box. In both cases, the slope is pulled down, which skews the calculated dimension when using L^2 . By comparing to the outlier-resistant Theil-Sen estimator, we were able to ascertain what appears to be a more robust scaling regime for the box dimension, which is $\epsilon \in [16, R/16]$ pixels. Restricting the sequence of up to 30 unique logarithmic intervals to this domain yields a statistically stable dimension metric, as demonstrated for a representative cluster in Figure 5.1.

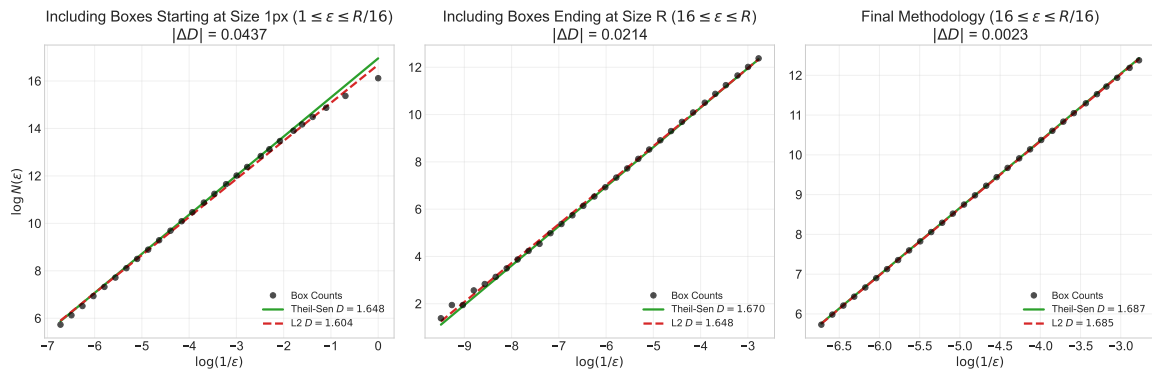


Figure 5.1: Comparison of Ordinary Least Squares (L^2) and Theil-Sen (TS) regressions for extracting the box-counting dimension of a single DLA cluster. To directly yield the fractal dimension D_0 as a positive gradient, the data is plotted as $\log N(\epsilon)$ against $\log(1/\epsilon)$. The annotated metric $\Delta D = |D_{L2} - D_{TS}|$ quantifies the discrepancy between the two statistical estimators. Including boxes that are too small or large relative to the cluster can have significant effects on the resulting dimension statistic.

5.2.2 Robustness of Growth Rate Dimension

As seen in Figure 5.1, upon adjusting the scaling, the two forms of regression produce near identical outputs. This was consistent across the random sample of clusters we compared, and additionally the direction of the bias was not skewed one way or another. In the case of the growth rate dimension, the discrepancy was again similarly negligible, with no directional bias.

When fitting the growth rate, we again had to adjust the start points, and the frequency of sampling, to ensure a robust fit. Again the Theil-Sen estimator was used as a point of comparison. After heuristic analysis we settled on a start point at $N=1000$ particles, which is large enough to avoid initial finite-size effects in the growth regime, and sampled 50 samples every power of 10.

From here on we use L^2 regression, having satisfied ourselves with the robustness of the slopes at the chosen scales.

Remark 5.3. Interestingly, when sampling evenly on a linear scale rather than logarithmic (biasing the fit towards the later stages of growth), this causes a more dramatic change in the resulting growth rate and corresponding dimension metric in the Lattice and Hybrid models. This aligns with our later discussion on anisotropy.

5.3 Results and Statistical Errors

Table 5.1: Summary of final dimensional metrics for 10^6 (1M) and 10^7 (10M) particle clusters across different simulation environments. Values are presented as the ensemble mean $\pm$ one standard deviation.

Model (Size)	Growth Rate (β)	Growth Dim (D_β)	Box Dim (D_{box})
Off-Lattice (1M)	0.583 ± 0.007	1.716 ± 0.020	1.667 ± 0.004
Off-Lattice (10M)	0.583 ± 0.004	1.714 ± 0.013	1.682 ± 0.002
Lattice (1M)	0.589 ± 0.007	1.697 ± 0.021	1.672 ± 0.004
Lattice (10M)	0.595 ± 0.006	1.682 ± 0.016	1.685 ± 0.002
Hybrid (1M)	0.592 ± 0.007	1.691 ± 0.021	1.672 ± 0.004
Hybrid (10M)	0.597 ± 0.005	1.674 ± 0.014	1.684 ± 0.002

The final results for both dimension metrics across the different models and cluster sizes are summarised in Table 5.1. We report the ensemble averages alongside the sample standard deviation (σ). In line with the DLA literature (e.g. [9]), σ is utilised as a direct, descriptive proxy for statistical uncertainty. This approach serves to capture the macroscopic inter-cluster variability inherent to the stochastic growth process. Notably, the inter-cluster variance for the box dimension is narrower than for the growth dimension. It would therefore appear to be a more stable metric, and less sensitive to the stochasticity of the cluster growing process at these scales.

Our recorded values for the kinetic growth dimension (D_β) closely align with the established literature. For the Off-Lattice model, our asymptotic result of 1.714 ± 0.013 matches the universally accepted consensus of $D \approx 1.71$ [23]. Similarly, our Lattice model result of 1.682 ± 0.016 at 10^7 particles aligns closely with Grebenkov and Belyaev’s result of 1.678 ± 0.011 , which was derived from slightly larger clusters of $\sim 5 \times 10^7$ particles [9]. While those authors also reported a lower box-counting dimension of 1.666 ± 0.004 , we attribute this discrepancy to the more restricted scaling regime applied to our analysis, justified in Section 5.2.

For our Hybrid model, we benchmark against the similar 4-antennae model presented by Menshutin [11], where their proposed logarithmic scaling relationship predicts dimension values of 1.694 and 1.653 for clusters of size 10^6 and 10^7 respectively. While our results follow a similar downward trajectory, direct quantitative comparisons must be treated with caution; as discussed in Chapter 3, we cannot be sure of algorithmic equivalence, and the methodologies used to compute the fractal dimensions also vary.

To visualise these macroscopic statistics, Figure 5.2 presents the probability density distributions of the two dimension metrics across the cluster ensembles for each model and size. The distributions reveal two critical observations: first, there is a distinct structural similarity between the results of the Hybrid and Lattice models; second, the two separate dimension statistics (D_β and D_{box}) do not perfectly align with one another. Investigating these two phenomena will form the remainder of this chapter.

5.4 Comparing Box-Dimension and Growth Rate

There is a small amount of literature that compares the box-counting dimension (D_{box}) of DLA to the proxy derived from the growth rate (D_β). [9] noted that for a sample of 100 lattice clusters of size $\approx 5 \times 10^7$ particles, the two statistics differ by under 1%. Other literature focusing on the box dimension includes [24], but they work with lattice clusters of size order

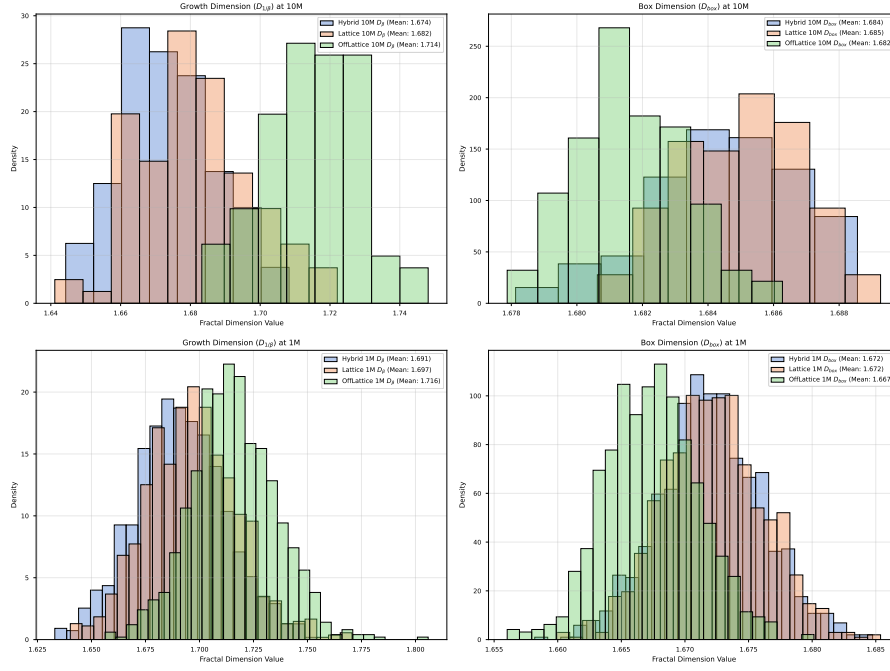


Figure 5.2: Probability density distribution of the Growth Dimension (D_β) and Box-Counting Dimension (D_{box}) for 10^6 (1M) and 10^7 (10M) clusters.

10^4 particles, which are susceptible to finite-size effects. We found little evidence of full-cluster box-counting applied to massive off-lattice aggregates, though implementations of the sandbox method (a distinct but related spatial scaling technique) on 10^6 particle Off-Lattice clusters estimate the dimension to be heavily depressed at $D \approx 1.685$ [25].

Our asymptotic dimension measurements at $N = 10^7$ particles, summarised in Table 5.1, reveal two distinct dimensional behaviors across the simulation spaces:

First, across all models and scales there is a noticeable depression in the box-counting dimension ($D_{box} \approx 1.68$) relative to the classical theoretical value of 1.71. One hypothesis is that this suppression is a finite-size artefact caused by evaluating the static bulk of the clusters. The growth dimension (D_β) is likely more influenced by the active outer region of the cluster, while the box-counting dimension averages over the entire structure. This aligns with the results of [24], where they performed box-counting on both the entirety and just the outer perimeter of the cluster, and found that the box dimension of the perimeter was much closer to the theoretical value of 1.71. However, they performed this analysis over smaller clusters ($N \approx 5 \times 10^4$). When we performed a similar analysis on a single Lattice cluster ($N = 10^7$), we found that approximately 96% of the cluster lay on the perimeter (i.e. had at least one vacant adjacent site), and the box dimension was unchanged. Indeed, even when performing a weighted box dimension calculation, matching [24], the dimension remained unchanged. For this particular cluster, it was 1.685 across all four combinations. This suggests that the persistent discrepancy is not a finite-size measurement artefact, but rather suggests these metrics capture fundamentally different properties. D_{box} measures the global, static spatial capacity of the aggregate, whereas D_β captures the dynamic growth. As our results show, these two properties need not align perfectly in finite macroscopic simulations.

Second, the growth dimension (D_β) exposes a much larger inter-model discrepancy than the box dimension. At $N = 10^7$, the D_{box} values across all three models are highly consistent; while the Off-Lattice estimate (1.682 ± 0.002) is nominally lower than the Lattice value (1.685 ± 0.002), the difference is marginal and the values remain within a single standard deviation of one another. Conversely, the Lattice and Hybrid models yield D_β values more than two standard deviations below the Off-Lattice baseline of 1.714.

A limitation of both our analysis so far and the wider literature, is comparing these quantities at specific cluster sizes. As it is impossible to know at what scale we truly no longer have finite size effects, a potentially more useful comparison is that of the evolution of D_{box} and D_β as the cluster grows. To our knowledge, mapping this continuous evolutionary comparison constitutes a novel contribution. This evolution is presented in Figure 5.3.

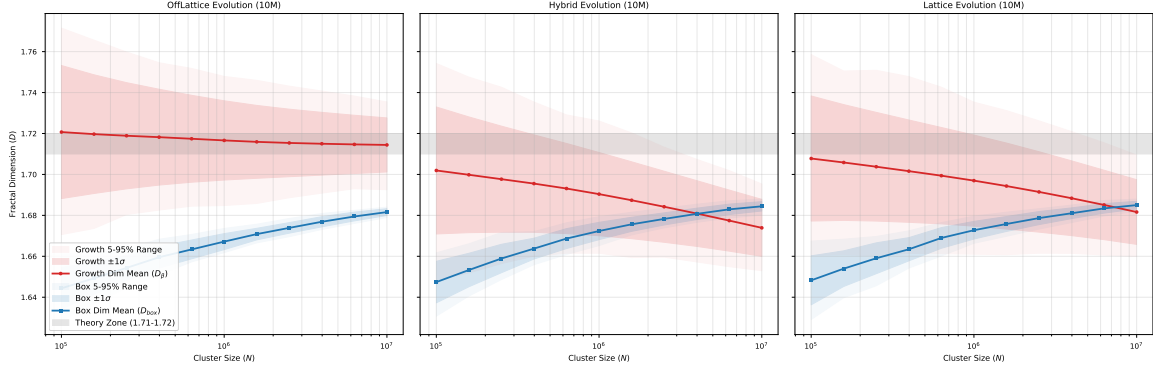


Figure 5.3: Statistical evolution of D_{box} and D_β against cluster size N for each model. Shaded regions represent regions of uncertainty.

As shown in Figure 5.3, the box dimension evolves similarly across all three models, while the growth dimension is clearly distinct in the Off-Lattice model from the other two models. This suggests that while $D_\beta = 1/\beta$ may be a valuable metric in and of itself, using it as a proxy for the dimension may in fact be misleading for finite sized clusters.

We theorise that for asymptotically large clusters, the two metrics may eventually converge in the Off-Lattice model, as the massive scale of the active outer shell begins to vastly outnumber the sparser inner core. While the trajectory in the left-hand panel of Figure 5.3 supports this hypothesis, providing definitive empirical evidence would require simulating clusters at scales currently beyond our computational limits. Conversely, the evolutionary trends for the Lattice and Hybrid models suggest such convergence is less likely, though finite-size constraints again prevent absolute conclusions.

Ultimately, it is crucial to recognize that a finite DLA simulation is not a true mathematical fractal, but rather a pre-fractal physical approximation. Consequently, no single finite-size spatial statistic can perfectly capture its theoretical dimension. While the box-counting dimension is the more rigorously accepted standard for quantifying spatial scaling, the widespread use of the growth rate as a proxy (D_β) introduces nuances that must be carefully managed. Nevertheless, the growth rate (β) and its inverse remain fundamentally important parameters for characterising the transport dynamics of the model. In alignment with established literature, we will continue to refer to D_β as a ‘dimension’ for the remainder of this work, whilst explicitly acknowledging the mathematical distinctions between these geometric and kinetic measurements.

5.5 Comparing the Dimension Across Models

Regardless of the inconsistencies across the two metrics and their cause, a key feature prevalent in both is the similarity between the Hybrid and Lattice models. This is in line with the visuals in Chapter 4, where the density maps appear almost identical. Looking at the specific numbers in Table 5.1, the dimension statistics are indeed very closely aligned.

This behavioural split is evident when analysing the growth dimension (D_β) at the maximum simulated aggregate size of $N = 10^7$ particles. As shown in Table 5.1, the Hybrid model yields a growth dimension of 1.674 ± 0.014 , which is not separable from the Lattice model’s 1.682 ± 0.016 .

The standard deviations of these two models heavily overlap, something which can be visually verified in Figure 5.2.

In contrast, the Off-Lattice model maintains a significantly higher growth dimension of 1.714 ± 0.013 . Using the standard deviations as a descriptive measure of spread, we observe a distinct separation between this model and the two grid-based models. The gap between the Off-Lattice mean and the Hybrid mean ($\Delta \approx 0.04$) is nearly three times larger than their respective standard deviations, supporting the claim that the lattice constraints enforce a physically distinct, accelerated radial expansion.

Crucially, this structural divergence is not merely a boundary effect localised to specific particle counts, but a consistent pattern in their evolution. Figure 5.3 illustrates the dynamic evolution of both D_{box} and D_β across the three models from $N = 10^5$ to 10^7 particles.

As seen in the leftmost panel, the Off-Lattice model's kinetic dimension (red line) remains stable throughout its entire growth history, sitting close to the theoretical ~ 1.71 target zone. However, the Lattice and Hybrid evolution plots (centre and right panels) exhibit very similar decay. Rather than maintaining a stable dimension, both models peel away from the theoretical baseline, their D_β values steadily falling as the clusters grow larger.

Furthermore, the continuous evolution plots reveal that the box dimensions (D_{box}) of the Lattice and Hybrid models share similar upward trends. Intriguingly, this consistency in D_{box} extends to the Off-Lattice model as well. Despite the stark divergence in their kinetic growth rates (D_β), all three models yield comparable box-counting dimensions ($D_{box} \approx 1.68$). This suggests that the static, space-filling capacity of the bulk aggregate is robust to the underlying spatial geometry, whereas the growth rate is more sensitive to lattice anisotropy. So far in all respects the Hybrid and Lattice models are very similar in our simulations, which is strong evidence for our hypothesis that the difference in aggregation is the primary driver of the structural differences between the different models, while the transport mechanism itself is not a significant factor.

Although this indistinguishability confirms the expected equivalence between the Hybrid and pure Lattice models as assumed by [11], it does not independently validate their subsequent claim that the dimension will eventually converge to Kesten's theoretical lower bound of 1.5. To reach that asymptotic conclusion, Menshutina and Schur employed noise-reduction techniques, requiring their antennae to be hit multiple times before aggregation occurred. Since introducing noise reduction fundamentally modifies the microscopic aggregation rules, it may skew the underlying harmonic measure and risks altering the universality class of the fractal [26]. While our own standard D_β trajectories do exhibit a clear downward trend, the observed deceleration of this decay makes stabilisation at a higher mean-field value just as mathematically plausible. Ultimately, extrapolating the true asymptotic limit of standard DLA from noise-reduced approximations remains speculative, and our empirical evidence at $N = 10^7$ particles is currently insufficient to definitively favour Kesten's limit over a higher stabilisation point.

Chapter 6

Anisotropy in DLA

So far we have restricted our analysis to the fractal (box-counting) dimension and the (somewhat) related growth rate. Now we turn to a different property to directly describe the symmetry of the clusters: anisotropy.

6.1 What is Anisotropy?

An aggregate is isotropic if its macroscopic properties (e.g. growth rate) are statistically invariant under rotation. Off-Lattice DLA is theoretically isotropic, as the underlying transport and aggregation rules do not favour any particular direction.

Anisotropy occurs when this rotational symmetry is broken, resulting in preferential growth along specific directions. Early investigations, including those by [27] and [21], demonstrated that simulating DLA on a discrete square lattice alters the continuous rotational symmetry. Even though the discrete random walk macroscopically approximates isotropic Brownian motion, the strictly localised, grid-restricted sticking rules force the microscopic branches to align orthogonally. Over millions of particle additions, these microscopic grid alignments compound, concentrating the harmonic measure towards the extreme tips of the primary Cartesian axes.

To quantify this phenomenon, we decompose our global radius of gyration (R_g) and growth rate (β) into angular components. By partitioning the cluster into discrete angular sectors θ , we can track the dynamic expansion of individual branches over time.

6.2 Angular Radius of Gyration

We follow [9] in utilising the angular radius of gyration rather than the maximum sector radius to quantify directional growth in order to mitigate the greater statistical noise associated with tracking single-particle boundary tips. The 2D plane is partitioned into n_s equal angular sectors S_θ originating from the centre of the cluster. For an aggregate of total mass N , the angular radius of gyration, $R_g(\theta, N)$, is given by:

$$R_g(\theta, N) = \left[\frac{1}{n_\theta(N)} \sum_{i=1}^N |\mathbf{r}_i|^2 \cdot \mathbb{I}(\mathbf{r}_i \in S_\theta) \right]^{1/2}$$

where $\mathbf{r}_i$ is the position vector of the i -th particle, $\mathbb{I}(\mathbf{r}_i \in S_\theta)$ is the indicator function (yielding 1 if the particle lies within sector S_θ and 0 otherwise), and $n_\theta(N)$ is the total number of particles occupying that specific sector at mass N .

We set the number of sectors to $n_s = 360$, corresponding to 1° angular wedges. We computed $R_g(\theta, N)$ at snapshots $N = 2^{10}$ to $N = 2^{23}$ particles for each of the 10^7 -particle clusters, and display the averages in Figure 6.1.

The Off-Lattice model exhibits a circular symmetry in the polar projection, with the semi-log plot showing a flat profile across all angles. In contrast, the Lattice and Hybrid models display a pronounced fourfold symmetry, with peaks at 0° , 90° , 180° , and 270° corresponding to the Cartesian axes, and troughs at 45° , 135° , 225° , and 315° corresponding to the diagonals. The semi-log plots also match this pattern. This is strong visual evidence of the anisotropic growth effects induced by the lattice constraints. The near identical profiles of the Lattice and Hybrid

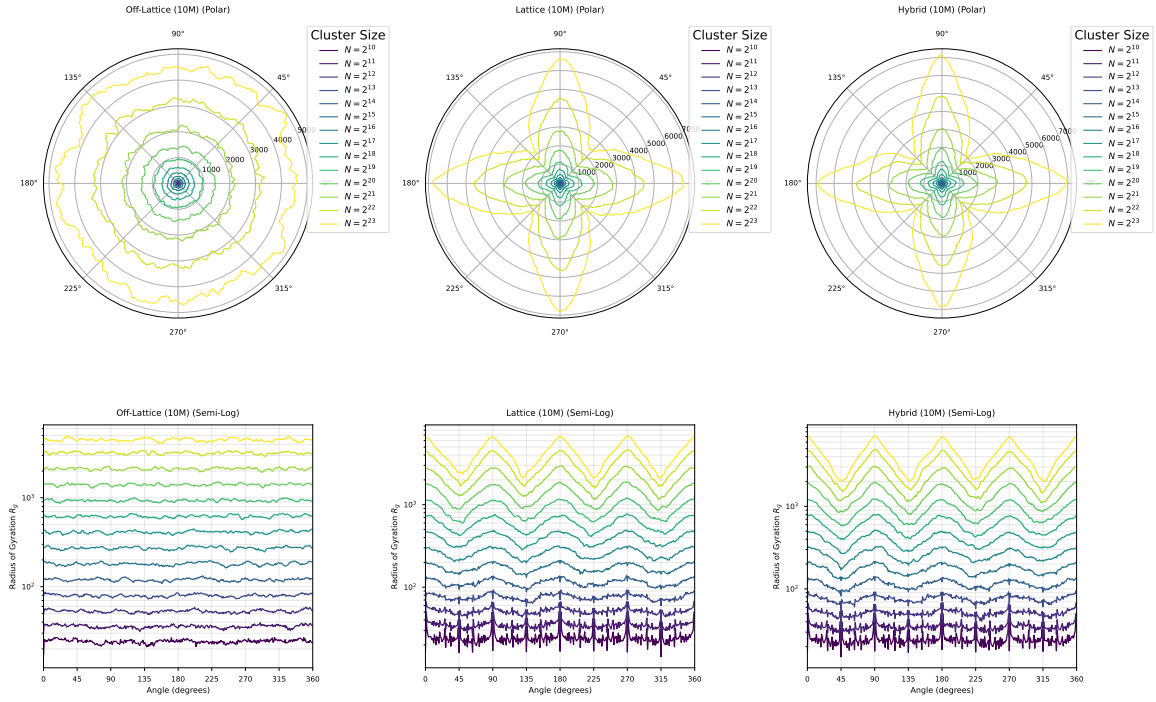


Figure 6.1: Evolution of the Angular Radius of Gyration, $R_g(\theta)$, at sequential mass snapshots ($N = 2^{10} \dots 2^{23}$) for the 10^7 -particle clusters, evaluated for 1° angular wedges. **Top Row:** Polar contour projections. **Bottom Row:** Semi-logarithmic projections unrolling $R_g(\theta)$ against the angle θ .

models at each snapshot are further evidence that the aggregation rule is the primary driver of this anisotropy.

6.3 Angular Growth Rate Analysis

While the angular radius of gyration effectively captures the spatial boundary at specific snapshots, it is ultimately an absolute distance metric. A likely more useful quantity for understanding the limiting behaviour is the angular growth rate, $\beta(\theta)$, which is both a dynamic and scale-invariant metric.

To reveal the differential growth along the orientational extremes, we aggregate the angular radii of gyration $R_g(\theta, N)$ for the four Cartesian axes ($0^\circ, 90^\circ, 180^\circ, 270^\circ$) to define $R_{axis}(N)$, and for the four diagonal directions ($45^\circ, 135^\circ, 225^\circ, 315^\circ$) to define $R_{diag}(N)$. Because the number of particles in any individual 1° sector is significantly smaller than the total cluster mass, local tip-growth fluctuations are high. To mitigate this statistical noise, we again follow the methodology of [9] by choosing relatively large aggregation sectors. We utilise four 11° wedges centred on the respective axes and diagonals, and average the aggregated radii over the ensemble of 10^7 -particle clusters. This is displayed in Figure 6.2.

As depicted in Figure 6.2, the isotropic Off-Lattice model exhibits parallel scaling between its axial and diagonal components. However, for the Lattice and Hybrid models, the log-log scaling laws diverge. Our growth rate numbers for the Lattice model are consistent with those of [9], where for 100 clusters of size $N \approx 5 \times 10^7$ particles, they found $\beta_{axis} = 0.612$ and $\beta_{diag} = 0.535$.

To compare the different models quantitatively, we consider the ratio of the two growth rates $A_R = \beta_{axis}/\beta_{diag}$. A number near 1 indicates isotropy, while a number significantly above 1 indicates anisotropy. For the Off-Lattice model, we find $A_R = 1.001 \pm 0.069$, confirming its isotropic nature. The Lattice and Hybrid models, however, yield $A_R = 1.217 \pm 0.084$ and $A_R = 1.245 \pm 0.090$ respectively, indicating a significant anisotropic bias towards the Cartesian

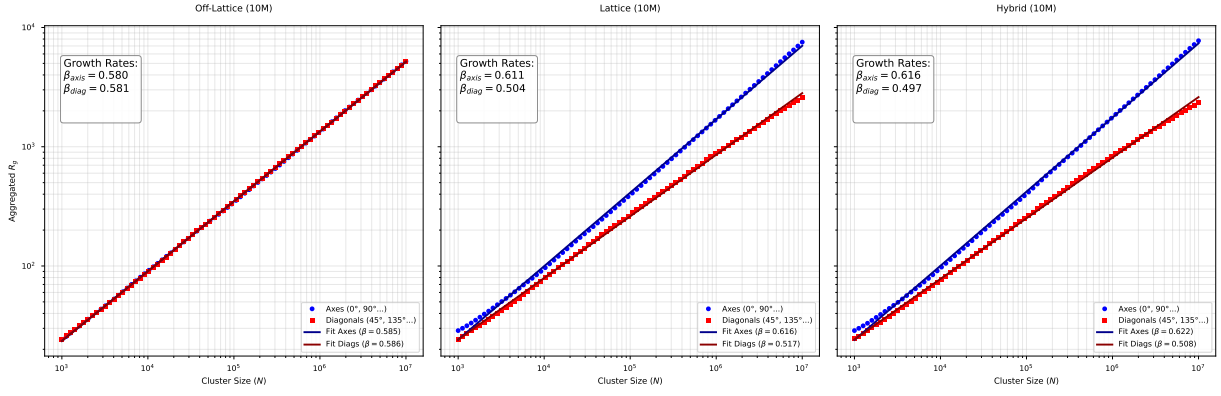


Figure 6.2: Aggregated angular radii of gyration for the cardinal axes, $R_{axis}(N)$ (blue circles), and the diagonals, $R_{diag}(N)$ (red squares). Values are calculated by averaging across four 11° directional wedges centred on their respective orientational angles. Radial snapshots were evaluated at logarithmically spaced mass intervals and averaged over the ensemble of 10^7 -particle clusters. L^2 regressions (solid lines) are performed on the stable scaling regime ($N \geq 10^3$) to extract the regional growth rate exponents, β_{axis} and β_{diag} .

axes.

The choice of a set of four 11° wedges is somewhat arbitrary, and chosen only to allow for a close comparison with [9]. We therefore introduce an additional statistic without this dependence.

Drawing inspiration from the multipole expansions utilised in [28], we define a mathematically analogous metric, the Fourier Score (C_4), adapted specifically for our angular growth rates. To reduce noise we first downsample our $\beta(\theta)$ profile into 90 uniform 4° sectors (centred on axes). We then compute the discrete Fourier transform, extracting the magnitude of the 4th harmonic (A_4) and normalising it by the 0th harmonic A_0 , i.e. $C_4 = \frac{|A_4|}{A_0}$.

An isotropic cluster theoretically yields $C_4 \approx 0$. As displayed in Table 6.1, the Off-Lattice model yields a score of $C_4 = 0.023 \pm 0.012$. In contrast, the Lattice and Hybrid models yield higher scores of 0.057 ± 0.016 and 0.062 ± 0.016 respectively. This is more evidence of the four-fold anisotropy governing the grid-constrained models, independent of the diffusion mechanism.

Table 6.1: Summary of angular anisotropy metrics for 10^6 (1M) and 10^7 (10M) particle clusters. Values are presented as the ensemble mean $\pm$ one standard deviation.

Model (Size)	Anisotropy Ratio (A_R)	Fourier Score (C_4)
Off-Lattice (1M)	1.010 ± 0.095	0.032 ± 0.016
Off-Lattice (10M)	1.001 ± 0.069	0.023 ± 0.012
Lattice (1M)	1.174 ± 0.112	0.055 ± 0.021
Lattice (10M)	1.217 ± 0.084	0.057 ± 0.016
Hybrid (1M)	1.189 ± 0.114	0.058 ± 0.021
Hybrid (10M)	1.245 ± 0.090	0.062 ± 0.016

To visualise the full angular profile of this divergence, we perform a coarse-grained angular growth analysis across the entire 360° spectrum. We group our existing 1° sector data into larger bins of 9° (40 sectors); an odd integer factor of 360 was specifically chosen to ensure the bins align symmetrically across the cardinal axes.

This is shown in Figure 6.3, where the angular growth rate $\beta(\theta)$ is plotted against the angle θ for each model. The green dashed line indicates the global mean growth rate across all angles.

There is a significant discrepancy between this mean value and the growth rates reported in Table 5.1 in the case of the two anisotropic models. This is because here each wedge is weighted equally, while the overall growth rate uses the radius of gyration of the whole cluster, where a large proportion of the particles are in the axial wedges, which have a higher growth rate. This is further evidence of the sensitivity of the growth dimension D_β to anisotropy.

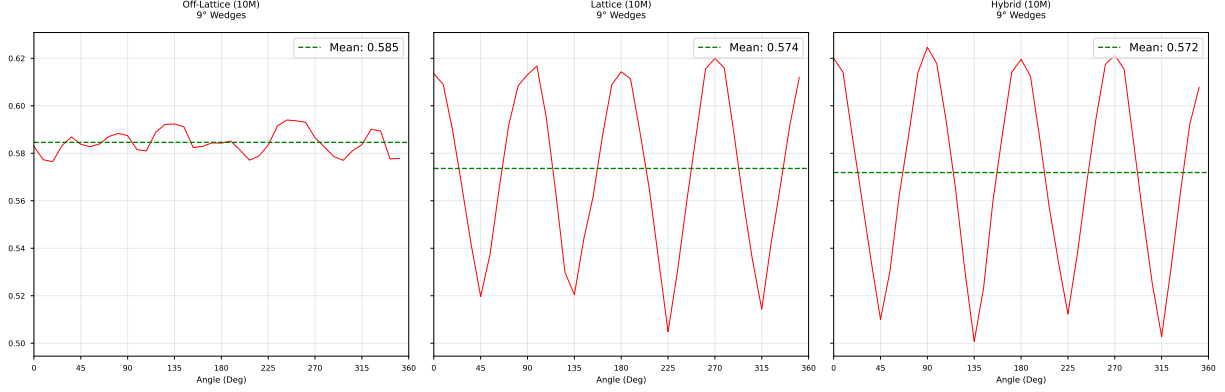


Figure 6.3: The angular growth rate $\beta(\theta)$ as a function of the angle θ , obtained from linear fits at log-log scale of $R_g(\theta, N)$ versus N . Data is evaluated using 9° angular wedges (40 sectors). The green dashed line indicates the global mean β .

As expected, the Off-Lattice angular growth rate fluctuates stochastically around the theoretical isotropic mean, with no dominant directional preference. The Lattice and Hybrid models, however, display periodic oscillations aligned with the Cartesian axes. Because the growth rate is an exponent ($R_g \sim N^\beta$), even a small fractional difference between β_{axes} and $\beta_{diagonals}$ results in an exponential separation in physical length.

When resolving the angular growth rate with highly granular bins (the angular wedges), a subtle but potentially important feature emerges in the Lattice and Hybrid models. Rather than peaking smoothly at 0° and 90° , the angular growth rate $\beta(\theta)$ exhibits more complex behaviour, which varies depending on the bin-sizing. We display this phenomenon in Figure 6.4, where we plot $\beta(\theta)$ for the Lattice model at three different bin sizes: 9° , 3° , and 1° wedges.

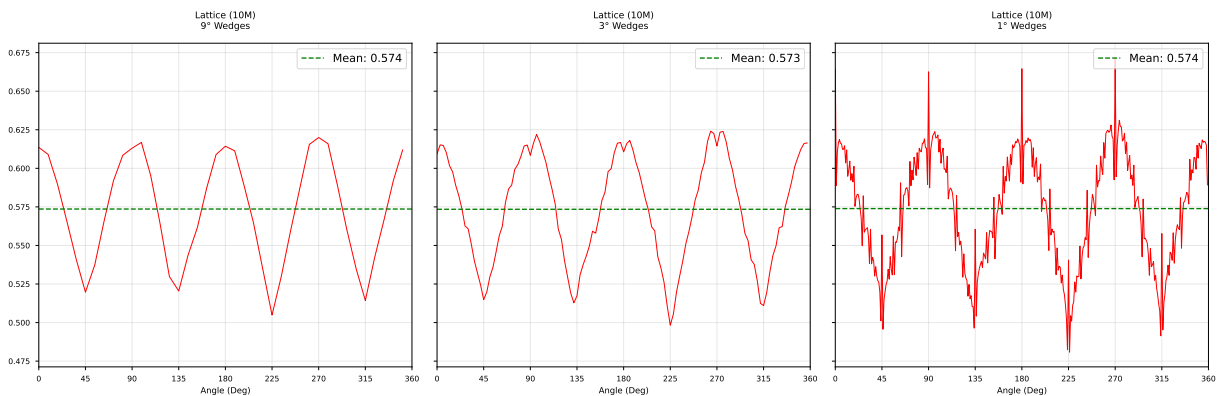


Figure 6.4: High-resolution angular growth rate $\beta(\theta)$ for the Lattice (10^7) model evaluated at progressively finer bin sizes (9° , 3° , and 1° wedges).

In Figure 6.4, the fine 1° binning reveals that there is a ‘W’-shaped profile along each axis, with an extreme spike exactly on axis accompanied by a significant dip in the adjacent bins. Looking at the 3° binning, the central spike is no longer visible, and the profile instead shows a slight dip on axis. We are not aware of a comparable observation in the literature reviewed here. Further binning results in a smoothing out of the entire profile in the 9° case, matching

the results in [9]. This illustrates how the choice of binning can have a significant effect on the resulting profile, and how the underlying mechanics can be obscured by coarse-graining.

Notably, this shadowing effect was observed in both the Lattice and Hybrid models on all four axes, suggesting it is not simply a product of stochastic noise, but again another characteristic of the lattice-influenced geometry. While the exact cause could warrant further investigation, its presence in both models is again further evidence of the structural equivalence between the two models.

Chapter 7

Conclusion

The primary objective of this dissertation was to investigate the mechanisms underlying the structural differences between discrete and continuous models of Diffusion-Limited Aggregation (DLA). To this end, we combined standard potential-theoretic ideas with accelerated simulation methods for both Lattice and Off-Lattice DLA, and used these to generate large statistical ensembles of aggregates, including 100 clusters of size 10^7 in each model. This provided a computational basis for a controlled comparison of macroscopic growth across the discrete and continuous settings.

The work presented here is not intended as a new theoretical foundation for DLA, but rather as a structured computational and analytical study built on existing mathematical and numerical frameworks. Within that framework, the dissertation makes several concrete contributions. First, it develops and implements a controlled Hybrid DLA model designed to separate, as far as possible, the role of the diffusion process from that of lattice-constrained aggregation geometry. Second, it uses a calibration procedure based on the Theil-Sen estimator to identify suitable fitting regimes for large-scale dimension estimates and to check that the production L^2 fits are not visibly dominated by boundary effects in the ranges considered. Third, it compares the growth-based dimension D_β with the box dimension D_{box} , providing evidence that these quantities need not coincide as finite-size descriptors of DLA clusters. Finally, it supplements the dimension analysis with angular growth statistics, which reveal a sharply localised reduction in growth immediately off the principal lattice axes in the grid-constrained models.

Taken together, these results support the view that, for the three models studied here and at the scales examined, the macroscopic anisotropy of square-lattice DLA is driven primarily by the aggregation geometry rather than by the transport mechanism alone. In particular, the Hybrid and Lattice models remain close to one another in both dimension estimates and angular growth statistics, while the Off-Lattice model behaves differently. This does not prove that transport effects are absent in general, nor does it rule out subtler differences emerging at larger scales or under different diagnostics. It does, however, indicate that the local geometric constraint imposed at the point of aggregation is the dominant source of the observed structural divergence in the present simulations.

Several directions for further work arise naturally from this study. The most immediate are:

- **Direct analysis of harmonic measure:** The simulation framework could be adapted to estimate the harmonic measure on the cluster boundary directly, for example by releasing non-aggregating probe particles and recording their hitting distribution. This would provide a more direct link between the probabilistic growth rule and the observed morphology.
- **A ‘Reverse’ Hybrid model:** A complementary model could be constructed in which transport is performed by a lattice random walk, but aggregation is treated continuously using exact adhesion. Comparing such a model with those studied here would provide a further test of the relative influence of transport and aggregation geometry.

Ultimately, this dissertation shows that even very simple local growth rules can produce subtle and model-dependent large-scale structure. By combining ideas from probability, potential theory, and large-scale computation, it provides further evidence that the geometry of aggregation plays a central role in shaping DLA clusters, while also highlighting the limits of what can be concluded from finite simulations alone.

Appendix A

Preliminaries to Potential Theory

A.1 Martingales and the Optional Stopping Theorem

In this section, we review the necessary tools in probability for linking random walks to potential theory. We begin by defining a probability space.

Definition A.1 (Probability Space). *A probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a triple where Ω is a sample set, $\mathcal{F}$ is a σ -algebra on Ω representing the set of events, and $\mathbb{P}: \mathcal{F} \rightarrow [0, 1]$ is a countably additive set function such that $\mathbb{P}(\Omega) = 1$.*

We now introduce filtrations, which formalise the idea of accumulating information over time.

Definition A.2 (Filtration). *A filtration on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a sequence of sub- σ -algebras $(\mathcal{F}_n)_{n \geq 0}$ such that:*

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots \subseteq \mathcal{F}.$$

The tuple $(\Omega, \mathcal{F}, (\mathcal{F}_n)_{n \geq 0}, \mathbb{P})$ is called a filtered probability space.

We now consider how decisions can be made at a given time based on the information available up to that time.

Definition A.3 (Stopping Time). *A random variable $\tau: \Omega \rightarrow \{0, 1, \dots, \infty\}$ is called a stopping time with respect to the filtration $(\mathcal{F}_n)_{n \geq 0}$ if for every $n \geq 0$:*

$$\{\omega \in \Omega : \tau(\omega) \leq n\} \in \mathcal{F}_n.$$

For example, ‘stop when the walker hits the cluster’ is a valid stopping time but ‘stop at the step before the walker hits the cluster’ is not.

With this in place, we model an unbiased process via the concept of a martingale.

Definition A.4 (Martingale). *A discrete-time stochastic process $(X_n)_{n \geq 0}$ is a martingale with respect to a filtration $(\mathcal{F}_n)_{n \geq 0}$ if:*

1. X_n is adapted to the filtration (i.e., X_n is $\mathcal{F}_n$ -measurable for all n),
2. $\mathbb{E}[|X_n|] < \infty$ for all n ,
3. $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n$ almost surely.

A martingale essentially encapsulates the idea of a ‘fair game’. The condition $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n$ implies that the best prediction for the next value is the current value, even with the entire history of the process.

A key result regarding martingales is the Optional Stopping Theorem. There are multiple versions, and we only state one here sufficient for our purposes.

Theorem A.5 (Doob’s Optional Stopping Theorem). *Let X be a martingale on a filtered probability space. If τ is a stopping time such that $\mathbb{P}(\tau < \infty) = 1$, and X is uniformly bounded (i.e., there exists K such that $|X_n(\omega)| \leq K$ for all n, ω), then:*

$$\mathbb{E}[X_\tau] = \mathbb{E}[X_0].$$

Proof. See Williams [29, Section 10.10]. □

This means that no stopping strategy τ can change the expected payout of a fair game. This will allow us to relate the value of a harmonic function at a starting point x (start of the walk) to its expected value at the boundary ∂D (end of the walk).

A.2 Measure Theory Concepts

In the study of potential theory, we will want to understand the relationship between the geometric length of a boundary (Lebesgue measure) and the probability that a random walker hits that boundary (harmonic measure). To rigorously define a probability density function linking these two concepts we require the notion of the Radon-Nikodym derivative.

We begin by defining absolute continuity between two measures, which is a necessary pre-requisite.

Remark A.6. Throughout this work, we assume our measurable spaces are equipped with the Borel σ -algebra.

Definition A.7 (Absolute Continuity). *Let $(\Omega, \mathcal{F})$ be a measurable space equipped with two measures μ and ν . We say that ν is absolutely continuous with respect to μ , denoted $\nu \ll \mu$, if for every $A \in \mathcal{F}$:*

$$\mu(A) = 0 \implies \nu(A) = 0.$$

Theorem A.8 (The Radon-Nikodym Theorem). *Let $(\Omega, \mathcal{F})$ be a measurable space, and let μ and ν be two finite measures such that $\nu \ll \mu$. Then, there exists a non-negative measurable function $f: \Omega \rightarrow [0, \infty)$, unique μ -almost everywhere, such that for any $A \in \mathcal{F}$:*

$$\nu(A) = \int_A f \, d\mu.$$

The function f is called the Radon-Nikodym derivative of ν with respect to μ and is denoted by:

$$f = \frac{d\nu}{d\mu}.$$

Proof. See Williams [29, Section 14.13]. □

Appendix B

Implementation Details and Optimisations

While Chapter 3 outlines the mathematical principles of the algorithms, the practical implementation requires specific data structures and optimisations to achieve the necessary scale. This appendix details the computational architecture of the simulation code, written in Python and accelerated using the Numba JIT (Just-In-Time) compiler. Recent literature has also explored the use of Python and Numba for DLA simulations [30]; however, such implementations rely on step-by-step discrete random walks. This native $\mathcal{O}(R^2)$ diffusive bottleneck restricts their feasible scale to clusters of roughly 10^4 particles. By contrast, the algorithmic accelerations detailed in Chapter 3 along with this appendix allow our framework to bypass this bottleneck, successfully generating datasets of 10^7 particles within a comparable computational timeframe.

The Lattice and Off-Lattice versions of the algorithm are versions of code created by Loh [15] and Bell [16] respectively. However, these were originally done in C and C++, and in translating to Python our implementation diverges both to account for differences in the language structure, and to account for the limitations of our computer (a 2017 Macbook with 8GB RAM). We will highlight some of these changes here, as well as explain some of the more programming-related concepts and structures behind the algorithms.

For additional background and explanations of the algorithms and data structures used, see [31].

B.1 Computational Framework

B.1.1 The Numba JIT Compiler

The simulations are implemented in Python to leverage its flexibility, and to be consistent with the analysis code. The core numerical routines are decorated with `@jit(fastmath=True)`. This compiles the Python bytecode into optimised LLVM machine code at runtime, allowing for performance parity with C++ while maintaining a modular, functional codebase. Our code is likely somewhat slower than the original implementations, but the JIT compiler mitigates this difference.

B.1.2 Rounding Semantics

One minor but important detail to consider in stochastic simulations is the rounding behaviour of floating-point co-ordinates, particularly in the Lattice and Hybrid models where the output is in integers.

- Python: Python 3 employs Banker’s Rounding (round half to even). For example, `round(2.5)` $\rightarrow$ 2 and `round(3.5)` $\rightarrow$ 4. This introduces a subtle statistical bias in random walks, as even coordinates are slightly favoured over odd ones.
- C and C++: The reference C++ implementations use the standard library `round()` function, which rounds half away from zero (e.g., `round(2.5)` $\rightarrow$ 3).

To ensure our random walks match the theoretical Lattice statistics and the reference datasets exactly, we implemented a custom kernel `_round_cpp(x)` that explicitly emulates the C++ behaviour in Python.

B.1.3 Memory Constraints and Parallel Execution

Generating statistically significant ensembles of clusters at $N = 10^7$ requires careful management of the fundamental trade-off between time complexity and space (memory) complexity, particularly given the our hardware limits (8GB RAM). Because each cluster represents an independent realisation of a stochastic process, ensemble generation is theoretically an embarrassingly parallel task; however, our execution strategy was strictly dictated by the differing memory footprints of the two models.

Lattice Model (Parallel Execution): The Lattice model possesses a highly compressed memory footprint due to its flattened bit-map architecture (see Appendix B.2). However, the reliance on pre-calculated finite jumps incurs a higher time complexity, requiring approximately 30-40 minutes to aggregate 10^7 particles. To optimise dataset generation, we exploited the low space complexity by deploying an asynchronous parallel execution pipeline via Python’s `multiprocessing` library. This allowed us to evaluate four independent clusters simultaneously across available CPU cores.

Off-Lattice Model (Sequential Execution): Conversely, the continuous Off-Lattice model is constrained by space complexity rather than time. Storing exact double-precision floating-point coordinates and maintaining the $\mathcal{O}(1)$ dynamic spatial hash significantly increases the baseline memory overhead. However, the massive algorithmic acceleration provided by the conformal mapping technique reduces the execution time to under 15 minutes for 10^7 particles. To prevent memory thrashing and out-of-memory (OOM) failures on our hardware, Off-Lattice realisations were generated strictly sequentially. The Hybrid model was treated in the same way.

B.2 Spatial Indexing and Hierarchical Skipping

In the diffusion steps of all versions of DLA, we require the distance from the walker to the nearest particle in the cluster to determine the maximum allowed step-size. Additionally, this distance is required in the Off-Lattice algorithm for the adhesion step, where we must identify the two nearest particles to evaluate the conformal map.

To compute this efficiently, we employ a hierarchical spatial map. The fundamental geometry consists of multiple overlaid grids. Level 0 represents the finest spatial resolution, while Level k consists of cells with physical dimensions $2^k \times 2^k$. Geometrically, a single cell at Level $k + 1$ exactly encompasses four underlying cells at Level k . By tracking the binary occupancy of these cells, we can rapidly determine the distance to the nearest particle without calculating $\mathcal{O}(N)$ pairwise distances.

B.2.1 Lattice: Flattened Bit-Map

Data Structure:

In the discrete model, the hierarchy must have enough layers to view individual lattice sites at the finest level. The reference C++ code uses a 2D array of pointers, with one array per hierarchy level. However, for large grids in Python, dynamic nested arrays introduce severe memory fragmentation and cache locality issues.

To resolve this, we implement a flattened 1D bit-map. All hierarchical levels are packed into a single contiguous `uint8` array, `flat_ss`, indexed via an offset table. We track occupancy strictly at the bit level (one bit per site) to minimise memory overhead.

Linear Search:

We utilise pre-computed Green’s function tables for jumps of radius 2^0 to 2^8 (corresponding to squares of sizes 2 to 512). The routine `_diffuse_variable_flat` performs a linear scan starting from the coarsest useful level (pixel size ≈ 256) down to the finest:

1. Start: Check level $L_{\text{start}} = \max(2, L_{\text{max}} - 9)$.
2. Check: We examine the 3×3 neighbourhood of cells at this level. If all 9 cells are empty, it is geometrically guaranteed that there is no cluster particle within a radius of $2^{L_{\text{start}}}$ from the walker. We immediately apply the maximum available jump ($R = 256$) and exit.
3. Repeat: If the neighbourhood is occupied, we decrement the level index to check finer resolutions ($R = 128, 64 \dots 1$).

Because lattice walkers have a capped maximum jump size due to the finite pre-calculated tables, they are frequently querying the top of this truncated hierarchy. Furthermore, since DLA walkers spend the vast majority of their time in empty voids, this coarsest 3×3 check usually succeeds immediately. Therefore, a linear scan is practically faster and computationally lighter than the alternatives in this specific case.

B.2.2 Off-Lattice: Dual-Structure Strategy

The continuous model separates the large-scale diffusion (Skipping) from the local adhesion (Hitting) into two distinct structures to optimise for differing algorithmic constraints.

B.2.2.1 The Coarse Hierarchy (Skipping)

Structure:

A dynamic list of 2D arrays. Unlike the Lattice version, we do not flatten this into a single array. At the finest level, the mesh width is approximately 24 units, which is sufficiently granular, meaning the memory pressure is orders of magnitude lower than the unit-resolution lattice grid.

Binary Search:

Since continuous walkers can jump any arbitrary distance, our hierarchy spans the entire simulation domain up to L_{max} . If a walker is deep inside a narrow fjord, it might be safe to jump only up to Level 3, but the hierarchy extends up to Level 15. A linear scan down from the top would waste numerous iterations. Instead, we use a binary search to locate the exact transition layer between ‘Occupied’ and ‘Empty’. This means the search time scales logarithmically with the number of levels, ensuring consistent, stable performance regardless of whether the walker is in a vast void or a constrained geometric boundary.

B.2.2.2 Deterministic Spatial Binning (Hitting)

Structure:

When the hierarchy indicates the walker is in the immediate vicinity of the cluster, we require the exact particle coordinates for the continuous adhesion step. However, the coarse hierarchy only records Boolean occupancy. To query nearest neighbours in $\mathcal{O}(1)$ time without dynamic memory allocation, we implement a static cell-linked list, functioning as a deterministic spatial binning structure.

The hash function is a simple geometric mapping: we obtain a 1D cell index by floor-dividing the continuous coordinates by the fine grid mesh size. This mimics a List of Lists using two pre-allocated integer arrays:

1. `grid_head[cell_index]`: Stores the integer index of the most recently added particle in that grid cell. Initialised to -1 .
2. `particle_next[particle_index]`: Stores the integer index of the next particle in the same cell, forming a linked list of particle indices.

Insertion:

When a new particle with integer index P_{new} is added to cell C , we perform a standard $\mathcal{O}(1)$ linked-list push-front operation:

```
particle_next[P_new] = grid_head[C]
grid_head[C] = P_new
```

Bell’s original Off-Lattice C++ implementation utilised standard vectors, allocating one dynamic vector per cell in the finest layer of the hierarchy. We prefer this linked-list approach as it avoids the continuous resizing of arrays or the allocation of new objects, which is a major performance bottleneck in Python. In both cases, we can simply iterate through the localized list to find the exact closest particle to the walker.

B.3 Rejection Sampling

B.3.1 Concept and Motivation

Several probability distributions required for the Lattice simulation possess Cumulative Distribution Functions (CDFs) that are computationally expensive or impossible to invert analytically.

A naive approach would be to discretise the domain and build a lookup table (numerical integration). However, this introduces systematic discretisation error. Instead, we employ rejection sampling. This method allows us to generate samples from a complex target distribution $f(x)$ using a simpler ‘proposal’ distribution $g(x)$, producing exact samples from $f(x)$ without any approximation error.

B.3.2 The Algorithm

Given a target density $f(x)$ and an envelope density $g(x)$ such that $f(x) \leq Mg(x)$ for all x (where M is a constant scaling factor), the algorithm proceeds as follows:

Algorithm 1: Rejection Sampling

```

1 Function SampleRejection( $f, g, M$ ):
2   while True do
3      $X \leftarrow \text{Sample}(g)$  ;           // Propose from envelope
4      $U \leftarrow \text{RandomFloat}(0, 1)$  ; // Verification variable
5     if  $U \leq \frac{f(X)}{M \cdot g(X)}$  then
6       return  $X$  ;                     // Accept and return
    // Else: Reject and loop again
```

Geometrically, this is equivalent to generating uniform random points under the curve $Mg(x)$ and keeping only those that fall under the curve $f(x)$. While the runtime is non-deterministic (dependent on the acceptance rate $1/M$), it is often computationally superior to numerical inversion because it avoids evaluating expensive special functions (like integrals of Bessel functions) for every rejected sample.

B.3.3 Application 1: The Fuzzy Annulus

To eliminate lattice aliasing, we launch particles from a radial distribution proportional to the Kaiser-Bessel function I_0 .

- $f(x)$: $I_0(\beta\sqrt{1-x^2})$.

- $g(x)$: A Gaussian distribution $\mathcal{N}(0, \sigma^2)$ that strictly bounds I_0 .

We sample candidate radii from the Gaussian (using standard library functions) and reject them based on the ratio of the Bessel weight to the Gaussian weight.

B.3.4 Application 2: Walk-to-Line

When a walker projects onto the bounding box face, we must sample the exit position on the line $y = 0$.

- $f(x)$: The exact discrete flux calculated via the Method of Images ($P_{lattice} \propto u(x, 1) - u(x, -1)$).
- $g(x)$: The Cauchy distribution (the analytical solution for continuous Brownian motion).

As established by Loh [15], the continuous flux serves as a valid envelope for the discrete flux. By setting the scaling constant $M = 1$, we ensure the condition $f(x) \leq Mg(x)$ holds, resulting in a computationally efficient sampling routine with a high acceptance rate.

B.4 Walker’s Alias Method

B.4.1 Motivation

For the Walk-to-Square algorithm, we must sample the exit location of a walker from the boundary of a square. This distribution is non-uniform (the exit probability is highest at the midpoints of the edges and lowest at the corners).

One approach would be to search a Cumulative Distribution Function (CDF). For a square of perimeter N , this requires a binary search, scaling as $\mathcal{O}(\log N)$. As the square size increases (e.g., $N = 2048$ for a 512×512 square), this $\log N$ cost inside the inner loop becomes a significant bottleneck.

To eliminate this overhead, we follow Loh and implement Walker’s Alias Method, which reduces the sampling complexity to $\mathcal{O}(1)$ —constant time, regardless of the square size.

Essentially the Alias Method works by reshaping the uneven probability histogram into a rectangular block.

Imagine the probability mass function as a set of vertical bars. We normalise them so the average height is 1. The method then slices the excess height from over-full bins (where $P > 1$) and moves these blocks to fill the gaps in the under-full bins (where $P < 1$).

The result is a table of size N where every bucket contains exactly the same total probability mass ($1/N$), but that mass is shared between at most two outcomes: the original occupant of that bucket, and the overflow from one other bin.

B.4.2 Implementation

We pre-compute two lookup tables for every square size $L = 2^k$:

- **Prob Table** (U_i): The threshold probability for keeping the original index i .
- **Alias Table** (K_i): The alternate index if the threshold check fails.

Sampling is then performed by first a discrete uniform random sample, and then a continuous uniform sample which is compared to the threshold value:

Algorithm 2: Alias Sampling (Constant Time)

```

1 Function SampleAlias( $U, K$ ):
2    $i \leftarrow \text{RandomInteger}(0, N - 1)$  ;           // Select a bucket uniformly
3    $y \leftarrow \text{RandomFloat}(0, 1)$  ;           // Select height within bucket
4   if  $y < U[i]$  then
5     return  $i$  ;                               // Return bucket owner
6   else
7     return  $K[i]$  ;                             // Return bucket alias

```

B.5 Numerical Approximations

B.5.1 Custom Bessel Functions

The Kaiser-Bessel launching distribution requires the Modified Bessel Function of the first kind, $I_0(x)$. The standard Python implementation, `scipy.special.i0`, is not supported in Numba’s `nopython` mode (as it links to C Fortran libraries not visible to the JIT compiler). We therefore implemented the polynomial approximation from [32] (Eq. 9.8.1) directly in Python/Numba. This ensures that the launching bias correction is applied efficiently without breaking the JIT compilation chain or requiring callbacks to the Python interpreter.

B.6 Usability and Data Management

To facilitate large-scale parameter sweeps and ensure consistency between the discrete and continuous datasets, we introduced two workflow improvements to the reference implementation.

B.6.1 Heuristic Grid Sizing

The spatial extent of a DLA cluster is stochastic and difficult to predict a priori. Loh’s code requires the user to manually define the grid dimension L_{max} . Underestimating this parameter leads to runtime failures (truncation), while overestimating it increases memory overhead.

To automate this, we implemented a heuristic initialisation step. Before memory allocation, we estimate the expected cluster radius R_{est} using the known fractal dimension $D_f \approx 1.71$ and apply a safety factor of 1.5 to account for the irregularity of the dendritic arms:

$$L_{max} = \lceil \log_2(1.5 \times 2.2 \times N^{1/1.71}) \rceil \quad (\text{B.1})$$

This ensures the simulation domain dynamically scales to accommodate the target particle count N without manual intervention. We adapted this formula from the one used in [16], and consequently our Off-Lattice code, but with adjustments based off of our own experimentation.

B.6.2 Standardised Coordinate Output

The internal state representations of the two models are fundamentally different: the Lattice model uses a boolean bit-map of occupancy, while the Off-Lattice model uses complex floating-point numbers with a particle radius $r = 1$ (effective diameter $d = 2$).

To enable unified analysis, we implemented a standardisation layer that normalises both outputs into an identical format: a standardised $(N, 2)$ array of dimensionless coordinates.

- **Lattice Conversion:** We convert the sparse bit-map into a list of integer coordinates representing the centres of the occupied sites, shifting the seed to the origin $(0, 0)$. We also ensure these are only the occupied cells, as Loh’s code includes the sticky sites as well.
- **Off-Lattice and Hybrid Normalisation:** We rescale the continuous coordinates by a factor of 0.5. Since the Off-Lattice physics uses $d = 2$ and the Lattice uses unit spacing $d = 1$, this normalisation ensures that spatial metrics (such as the Radius of Gyration) are directly comparable across models.

This post-processing step allows us to apply the exact same analysis scripts (fractal dimension estimation, radial density functions) to all datasets agnostically, eliminating potential errors arising from manual unit conversion.

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